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2 Versions

AttriSelect: Efficient Attribute-Driven Data Selection for Targeted Vision Model Repair

(/pdf?id=JxkGmySuLb)

Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2), RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2)

01 May 2026 (modified: 15 Sept 2026) NeurIPS 2026 Conference Withdrawn Submission
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers, Authors Revisions (/revisions?id=JxkGmySuLb) BibTeX
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TL;DR: Attribute-Driven Data-Centric Bug Discovery and Repair

Abstract:
Targeted vision repair aims to correct model failures on rare semantic slices after a debugging pipeline identifies the failure slice and a generator produces corresponding repair images. In most existing workflows, these repair images are appended to the original training set, and the model is retrained or fine-tuned using all source data. While this preserves broad coverage, it also makes each repair cycle computationally expensive and time-consuming, especially as repairs accumulate over multiple iterations. Moreover, it provides little control over which source examples actually contribute to the repair. This poses a data selection problem related to active learning: rather than using the full dataset, the goal is to identify a compact yet informative subset that best supports repair while substantially reducing retraining cost. However, an effective repair subset must satisfy requirements beyond standard sample efficiency. In particular, it should preserve class coverage, retain rare repair-relevant attributes, and avoid redundancy with the generated repair data. Existing active learning methods typically prioritize uncertainty, diversity, or gradient-space coverage, but these signals are often poorly aligned with the semantic attributes that define the failure slice. To address this limitation, we propose AttriSelect, an attribute-driven selector that prioritizes samples based on attribute rarity, generated-data coverage, and soft class balance. We provide theoretical analysis and evaluate AttriSelect across image classification, pose estimation, and object detection. Across tasks and repair budgets, AttriSelect gives the strongest repair results in most settings, showing that explicit attribute metadata provides a repair-specific selection signal distinct from traditional active learning criteria.

Checklist Confirmation: I confirm that I have included a paper checklist in the paper PDF.
Supplementary Material: zip (/attachment?id=JxkGmySuLb&name=supplementary_material)
Responsible Reviewing: We acknowledge the responsible reviewing obligations as authors.
Primary Area: Data-centric aspects of AI (e.g., data selection, active learning, curriculum learning, online learning)
Secondary Area: Computer vision (e.g., object detection, segmentation, image classification, image generation, video)
Contribution Type: General: Most submissions will fall into this type.
Academic Integrity: I acknowledge that I have read the NeurIPS Handbook and commit to adhering to all policies in the Handbook (https://neurips.cc/Conferences/2026/MainTrackHandbook (https://neurips.cc/Conferences/2026/MainTrackHandbook)), the NeurIPS Code of Conduct and the NeurIPS Academic Integrity Policy.
LLM Usage: Not used at all (you can then skip the rest)
Declaration: I confirm that the above information is accurate.
Reviewer Nomination: Ouyang Xu (/profile?id=~Ouyang_Xu1)
Submission Number: 12417

Discussion (?id=JxkGmySuLb#discussion)

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Withdrawal by Authors

Withdrawal
by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))
15 Sept 2026, 22:35 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers, Authors

Withdrawal Confirmation: I have read and agree with the venue's withdrawal policy on behalf of myself and my co-authors.

Citation Checker by Program Chairs

Citation Checker by Program Chairs 04 Aug 2026, 13:22 (modified: 05 Sept 2026, 00:13)

Program Chairs, Senior Area Chairs, Area Chairs, Authors Revisions (/revisions?id=gpkMWKyq3J)

Comment:

Automatically flagged references

Here we list the references that were automatically flagged by our checker. Note that our checker may miss many hallucinated references, and some of the listed references may have been flagged erroneously. **It is therefore crucial that you check them manually, following the guidance above. We rely on the Area Chairs to decide whether these constitute violations of responsible authorship.**

For papers falling into either of the following categories, we expect your default recommendation to be a rejection on the grounds of irresponsible authorship. This is strong presumption rather than a strict rule; as the Area Chair, you retain discretion, and may recommend otherwise where the specific circumstances of a paper warrant it:

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Please complete the form “Hallucinated References Investigation” to communicate the results of your investigation prior to completing your meta reviews.

Citation Checker Report

Reference Check Summary

4/46 problematic references.

List of Problematic References with Reasoning

```
@inproceedings{chen2023safefix,
  author = {Muxi Chen and Yu Li and Qiang Xu},
  title = {Safefix: On human-interpretable model debug},
  booktitle = {Advances in Neural Information Processing Systems},
  year = {2023}
}
```

The paper title provided in the citation, 'Safefix: On human-interpretable model debug', is hallucinated. The actual paper authored by Muxi Chen, Yu Li, and Qiang Xu published in NeurIPS 2023 (Advances in Neural Information Processing Systems) is titled 'HiBug: On Human-Interpretable Model Debug'.

```
@article{chen2025safefix2,
  title={Safefix2: Efficient and interpretable error slice discovery for comprehensive model debugging},
  author={Chen, Muxi and Zhao, Chenchen and Xu, Qiang},
  journal={arXiv preprint arXiv:2501.16751},
  year={2025}
}
```

The paper associated with arXiv ID 2501.16751 and authors Muxi Chen, Chenchen Zhao, and Qiang Xu is titled 'HiBug2: Efficient and Interpretable Error Slice Discovery for Comprehensive Model Debugging', not 'Safefix2: Efficient and interpretable error slice discovery for comprehensive model debugging'.

```
@article{li2024your,
  author = {X Li et al.},
  title = {Your vision-language model itself is a strong filter},
  journal = {arXiv preprint arXiv:2402.12501},
  year = {2024}
}
```

The paper with arXiv ID 2402.12501 and title 'Your Vision-Language Model Itself Is a Strong Filter: Towards High-Quality Instruction Tuning with Data Selection' exists, but the authors are completely mismatched. The actual first author is Ruibo Chen (along with Yihan Wu, Lichang Chen, Guodong Liu, Qi He, Tianyi Xiong, Chenxi Liu, Junfeng Guo, and Heng Huang), not 'X Li'.

```
@article{zhang2023adding,
  author = {Lvmin Zhang and Maneesh Wu and Chengyue Zhang and Yijun He and Siwei Zhang and Xiaolong Xu and Jingwan Yang and Kevin Shih and Xintao Wang and Yingying Zhang and others},
  title = {Adding conditional control to text-to-image diffusion models},
  journal = {arXiv preprint arXiv:2302.05543},
  year = {2023}
}
```

The citation heavily hallucinates the author list. The actual paper 'Adding Conditional Control to Text-to-Image Diffusion Models' (arXiv:2302.05543) was authored by only three people: Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. The provided BibTeX lists ten specific authors, many of whom are entirely fabricated (such as Maneesh Wu, Chengyue Zhang, Yijun He, Siwei Zhang, Xiaolong Xu, Jingwan Yang, Kevin Shih, Xintao Wang, and Yingying Zhang).

Hallucinated/problematic reference criteria

These are adopted from ICLR and ICML 2026 hallucinated reference guidance:

- If you are unable to find any document or publication with a title anywhere near the one provided in the reference, it should be considered hallucinated. Minor differences in title should not be considered evidence of a hallucinated reference.
- If the reference's authors are only slightly incorrect, e.g., a few subtle misspellings, missing a few authors, or wrong authorship order, we do not consider this to imply a hallucinated reference, but please let the submission's authors know in your meta-review that they should fix this issue.
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- If there is no evidence that the reference's venue exists, the reference should be considered hallucinated.
- Finally, we note that there are many generic references that refer to existing artifacts (e.g. "GPT-4o, OpenAI et al. 2024") but do not refer to any real document. These are fine.
- Even with the above guidelines in mind, there will likely be some borderline cases. Please bring those up with your SAC if you need additional guidance.

Official Review of Submission12417 by Reviewer yyPZ

Official Review by Reviewer yyPZ 25 Jun 2026, 01:34 (modified: 04 Aug 2026, 08:23)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ Revisions (/revisions?id=gFhG8tCROU)

Summary:
The paper introduces AttriSelect, a targeted data selection algorithm designed to efficiently repair semantic failure slices in computer vision models. Rather than relying on traditional active learning metrics like uncertainty, gradient embeddings, or geometric coverage, AttriSelect scores source training data using structural attribute metadata assigned by Vision-Language Models (VLMs).

The scoring function balances three components: class-conditional attribute rarity, a penalty for attribute values already abundant in the generated synthetic repair data, and a soft class-balancing term.

Contribution Type: General: Most submissions will fall into this type.
Strengths And Weaknesses:
Strengths:

1. Framing targeted model repair as an attribute-aware source selection problem is a highly practical perspective.
2. The methodology is straightforward.
3. Writing is clear.

Weakness:

1. Misleading Claims Regarding Computational Efficiency. The authors highlight in Table 2 that AttriSelect takes less than 2 seconds, claiming a speedup of up to 283× over baselines. However, this entirely omits the massive computational overhead of the prerequisite step: running Qwen3-VL to annotate the entire source pool. Appendix A states it takes 10–20 minutes to annotate just 1,000 images. For datasets like CelebA (80,000 images) or COCO (20,000 images), this requires tens to hundreds of GPU hours. Claiming high efficiency while ignoring the cost of generating the metadata that the algorithm relies upon is highly misleading.
2. Comparison look Unfair. AttriSelect is given privileged information like VLM-assigned attributes and explicit bug definitions, but the compared active learning baselines (Entropy, BADGE, CoreSet) do not have access to. The authors should compare it against a baseline with the same information, for example, an "Attribute-Filtered Random" or "Attribute-Filtered Entropy" baseline.
3. Despite the heavy pipeline (VLM tagging, diffusion generation, subset selection), the absolute Fix Rates (FR) in Table 1 remain quite low, mostly ranging between 5% and 13%.

Quality: 2: not good
Clarity: 3: good
Significance: 2: not good
Originality: 3: good

Questions:
See weaknesses

Limitations:
yes

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
NA

Code Of Conduct Acknowledgement: Yes
Responsible Reviewing Acknowledgement: Yes

Response to Reviewer yyPZ (Part 1)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

29 Jul 2026, 22:01 (modified: 30 Jul 2026, 15:45)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ

Revisions (/revisions?id=OEOqdUP4Km)

Comment:
Dear Reviewer yyPZ, thank you for raising these questions about end-to-end cost, comparison fairness, and the interpretation of FR.

This response is organized into four parts.

Q1 — End-to-End Computational Cost

Accounting Scope

Response. Table 2 of the submitted paper measures only source-subset selection time, not the complete repair pipeline, and the revision will make that scope explicit. The reported speedups apply to source selection after the attribute records are available. **They should not be interpreted as end-to-end speedups** over source annotation, repair-data construction, retraining, and evaluation.

We report repair-image generation separately because **the generated repair set is a fixed, publicly available dataset shared by every selector**, not a set regenerated for each method. Every selector receives the same repair images as pre-existing augmentation data. Regenerating the SafeFix [1] set would add the same one-time dataset-construction cost to every method without changing their runtime ordering. The table columns therefore cover source-side preprocessing and selection through retraining and evaluation. By contrast, **we include VLM source-pool assignment for AttriSelect** because it is a method-specific prerequisite.

Measured Source-Pool Assignment Time

To answer the assignment-cost question directly, we measured complete source-pool VLM assignment on one NVIDIA H200 GPU:

Dataset	Source Images	Assigned Attributes	H200 Assignment Time	Normalized Assignment Time (min / 1K images / attribute)
KITTI	5,000	6	32.77 min	1.09
ImageNet10	10,400	3	29.35 min	0.94
COCO	20,000	5	90 min	0.90

Normalized Assignment Time divides the total H200 assignment time by the source-pool size in thousands and the number of assigned attributes. For example, the COCO value is 90 / (20 x 5) = 0.90 minutes per 1,000 images per attribute.

Conservative Comparison

For a fixed task-level attribute vocabulary, **source-pool assignment is a one-time preprocessing cost, and its metadata can be reused** across source ratios, selectors, and later repair iterations on the same task. The full pool does not need to be annotated again for every repair. Here, **the first repair run assumes that source attributes must be assigned from scratch**. Its cost covers source preprocessing and selection through retraining and evaluation; repair-image generation is reported separately. Even so, **every AttriSelect row below charges the full fresh-assignment cost**. On one NVIDIA H200 GPU, the mean over two complete 5,000-image KITTI runs is **32.77 minutes**. The earlier estimate of 10--20 minutes per 1,000 images in the submitted paper was measured on an NVIDIA A5000, as stated in the appendix; on the H200, assignment takes several minutes per 1,000 images, depending on the number of required attributes. Assigning the complete 20,000-image COCO source pool on the same H200 takes approximately **1.50 hours**.

KITTI: 5,000 source images. Each AttriSelect row charges the mean fresh 5,000-image H200 VLM assignment time of 32.77 minutes.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours
0.1	42.17 min	42.18 min	48.82 min	1.37 h	1.12 h	43.47 min	43.53 min	1.01 h

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours
0.3	43.84 min	43.85 min	50.28 min	1.44 h	1.20 h	45.08 min	45.63 min	1.09 h
0.5	45.18 min	45.18 min	51.73 min	1.49 h	1.27 h	46.42 min	47.47 min	1.11 h
0.7	47.04 min	47.05 min	53.46 min	1.56 h	1.35 h	48.16 min	49.73 min	1.13 h
0.9	48.86 min	48.87 min	55.53 min	1.63 h	1.42 h	50.34 min	51.46 min	1.15 h

COCO: 20,000 source images. Image generation and generated-repair-image assignment are excluded. Each entry includes the applicable source preprocessing and selection cost, retraining, and evaluation. The AttriSelect column charges a fresh full-pool VLM assignment in every row.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours
0.1	4.41 min	4.42 min	10.98 min	16.18 min	19.63 min	5.08 min	6.97 min	1.57 h
0.3	8.18 min	8.18 min	14.74 min	39.86 min	49.76 min	8.93 min	17.86 min	1.62 h
0.5	12.37 min	12.36 min	19.04 min	1.08 h	1.56 h	13.08 min	29.47 min	1.66 h
0.7	15.73 min	15.75 min	22.27 min	1.45 h	2.29 h	16.34 min	37.63 min	1.69 h
0.9	19.66 min	19.65 min	26.28 min	1.87 h	3.01 h	20.18 min	40.94 min	1.72 h

Replying to Response to Reviewer yyPZ (Part 1)

Response to Reviewer yyPZ (Part 2)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

29 Jul 2026, 22:02 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ

Comment:

Q1 — End-to-End Computational Cost (continued)

Cost–Performance Interpretation

> **Cost takeaway.** Combined with the main results in the submitted paper, these tables show a trade-off between computational cost and repair performance. **Random, Stratified, UHerding, and PPAL are faster, especially at small source ratios, but their repair performance is generally lower.** For example, at COCO ratio 0.1, AttriSelect takes 1.57 hours when fresh annotation is charged, whereas Random, UHerding, and PPAL take 4.41, 5.08, and 6.97 minutes, respectively; their test AP values are 0.6048, 0.6013, and 0.5950, compared with 0.6183 for AttriSelect.
> **CoreSet and BADGE are generally the two strongest baselines besides AttriSelect, but they become much slower at high source ratios.** At ratio 0.9, they are even slower than AttriSelect after the AttriSelect total conservatively includes VLM annotation of the entire source pool. On COCO, AttriSelect takes 1.72 hours and reaches 0.6355 AP, compared with 1.87 hours/0.6202 for CoreSet and 3.01 hours/0.6213 for BADGE. On KITTI, AttriSelect takes 1.15 hours and reaches 0.5850 mAP50--95, compared with 1.63 hours/0.5781 for CoreSet and 1.42 hours/0.5810 for BADGE. Thus, **at ratio 0.9 on both COCO and KITTI, AttriSelect provides both better repair performance and lower wall-clock cost on the first repair run, even after charging full-pool VLM assignment**.

CoreSet [3] and BADGE [4] become substantially slower at high source ratios and on larger source pools because their standard greedy procedures update the distance from every one of the N candidates to each new center. Selecting $B = rN$ samples requires $O(NB) = O(rN^2)$ distance-update work at a fixed ratio. The measurements follow this scaling pattern: cost rises sharply with the retained ratio and grows further with the source pool. BADGE can be slower because its gradient embeddings are usually higher-dimensional. AttriSelect does not repeat candidate-to-center computations after its reusable attribute records are available.

Existing Metadata and Later Repair Cycles

Source attributes also need not be assigned from scratch when suitable metadata already exist. For example, the official CelebA release (https://mmlab.ie.cuhk.edu.hk/projects/CelebA.html (https://mmlab.ie.cuhk.edu.hk/projects/CelebA.html)) provides 40 binary attributes for each of its 202,599 images. **If these public annotations cover the required task-level vocabulary, AttriSelect can use them directly and incurs no VLM assignment cost.** New assignment is needed only for attributes missing from the available metadata.

Cost After Metadata Reuse

The tables are deliberately conservative for AttriSelect because they charge source assignment separately at every ratio. **In repeated repair cycles, the metadata are reused, so the annotation cost is amortized and the remaining online AttriSelect computation is below two seconds.** We will revise Table 2 to label its values explicitly as source-subset selector runtime, add this end-to-end decomposition, and clarify that **the reported 39--283x speedups apply only to source-subset selection after attribute data are available**.

Response to Reviewer yyPZ (Part 3)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

29 Jul 2026, 22:03 (modified: 29 Jul 2026, 22:03)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ

Revisions (/revisions?id=hRlVhQ5ggp)

[Deleted]

Replying to Response to Reviewer yyPZ (Part 1)

Response to Reviewer yyPZ (Part 3)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

29 Jul 2026, 22:04 (modified: 30 Jul 2026, 15:46)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ

Revisions (/revisions?id=r4UQKe5wbF)

Comment:

Q2 — Same-Information Attribute-Aware Baselines

Response. The generic active-learning baselines do not receive the attribute information used by AttriSelect. They test whether repair-specific metadata adds signal beyond uncertainty, geometric coverage, and gradient embeddings, but they do not test whether our score uses that metadata better than a simple attribute-aware rule.

On ImageNet10, the target slice is defined by "color=pink" and "texture=fabric". We added three baselines that receive the same attribute metadata and target definition:

- **Target-Slice + Stratified:** select exact pink-and-fabric source examples first and fill the remaining budget with class-stratified background samples;
- **Attribute-Filtered Random:** prioritize the pink-or-fabric source pool and sample randomly within class-balanced queues before filling from the background;
- **Attribute-Filtered Entropy:** use the same candidate pool and class-balanced queues, but rank candidates by predictive entropy from the fixed diagnosis checkpoint.

All methods select only source images and use the same fixed 1,000-image generated set, source budget, VLM annotations, class labels, diagnosis checkpoint, generated seed, and 15-epoch repair schedule. The table reports mean overall ACC and standard deviation over seeds 0, 42, 1234, 3407, and 2024. The AttriSelect values are the five-seed results reported in the appendix of the submitted paper.

Five-seed ImageNet10 same-information repair comparison. Metric: test accuracy (fraction; mean ± standard deviation).

Source ratio	Target-Slice + Stratified	Attribute-Filtered Random	Attribute-Filtered Entropy	AttriSelect
0.1	0.7303 ± 0.0051	0.7340 ± 0.0034	0.7317 ± 0.0025	**0.7369 ± 0.0043**
0.3	0.7385 ± 0.0024	0.7382 ± 0.0042	0.7365 ± 0.0023	**0.7492 ± 0.0044**
0.5	0.7438 ± 0.0037	0.7442 ± 0.0038	0.7442 ± 0.0031	**0.7508 ± 0.0010**
0.7	0.7434 ± 0.0065	0.7474 ± 0.0038	0.7482 ± 0.0020	**0.7515 ± 0.0037**
0.9	0.7485 ± 0.0026	0.7469 ± 0.0036	0.7503 ± 0.0045	**0.7562 ± 0.0019**

> **Key result.** Even when every baseline receives the same attribute metadata and target-slice definition, AttriSelect has the highest mean accuracy at all five source ratios. Its margin over the strongest same-information baseline ranges from 0.29 to 1.07 accuracy points, with an average of 0.59 points. This consistent but moderate advantage shows that the selection score provides additional value beyond access to the attributes alone.

Q3 — Interpreting Fix Rate and the Repair Budget

> "Despite the heavy pipeline (VLM tagging, diffusion generation, subset selection), the absolute Fix Rates (FR) in Table 1 remain quite low, mostly ranging between 5% and 13%."

Raw-Metric Interpretation

Response. Fix Rate (FR) should be read together with the original task metric. FR is the **Relative Error Reduction** toward the maximum task score, not the percentage of test examples or failure slices fixed. At the reported ratio-0.9 AttriSelect settings, the raw changes are:

- ImageNet10 accuracy: **0.7173 to 0.7562, or +3.89 percentage points**;
- CelebA accuracy: **0.9057 to 0.9196, or +1.39 percentage points**;
- COCO keypoint AP: **0.5902 to 0.6355, or +4.53 AP**;
- KITTI mAP50--95: **0.5626 to 0.5850, or +2.24 mAP**.

The slice-level heatmaps in Figures 2 and 4 of the submitted paper provide a more direct view of the localized repair effect. At source ratio 0.5, the largest improvements occur on the diagnosed values---pink and fabric for ImageNet10, and brown and rainy for COCO---while the remaining attribute slices show only small changes and no systematic degradation. Thus, the overall metric can understate the targeted benefit: AttriSelect improves the diagnosed failure slices without materially reducing performance on unrelated slices.

The reported 5--13% range represents a reduction in the model's remaining error, not a 5--13 percentage-point change in the raw task metric. AttriSelect already achieves **6.93% Relative Error Reduction at ImageNet10 ratio 0.1**, raising ACC from 0.7173 to 0.7369 (+1.96 percentage points) while selecting only 10% of the source pool. This is the highest FR at ratio 0.1: the strongest baseline, UHerding, reaches 5.59% FR and 0.7331 ACC. AttriSelect improves FR by **1.34 points (24.0% relative)** and ACC by **0.38 percentage points** under the same budget. Ratio 0.1 is also the most data- and retraining-efficient repair setting evaluated: it uses one-ninth as many selected source examples as ratio 0.9 while retaining the same generated repair set. The fixed source annotation and image-generation costs are accounted for separately in the end-to-end analysis.

Replying to Response to Reviewer yyPZ (Part 3)

Response to Reviewer yyPZ (Part 4)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

29 Jul 2026, 22:05 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ

Comment:

Q3 — Interpreting Fix Rate and the Repair Budget (continued)

Epoch-Budget Sensitivity

As specified in the appendix, the ImageNet10 diagnosis model is trained for 30 epochs at learning rate 1×10^{-4}, while repair uses 15 epochs at 1×10^{-5}. For CelebA, COCO, and KITTI, repair likewise **uses half as many epochs as diagnosis**, with learning rate 1×10^{-5}. **This moderate schedule makes the experiment a controlled comparison of source-selection methods within model repair. Because AttriSelect is a data-selection method, we compare it with different AL baselines under a matched repair budget instead of prolonging training to maximize performance on the underlying computer vision task.** Every selector uses the same number of repair epochs, learning rate, generated set, and source-data budget.

To test whether the reported FR values are mainly limited by this schedule, we also vary ImageNet10 repair training from 15 to 60 epochs, corresponding to $0.5\times$--$2\times$ the 30-epoch diagnosis schedule. We evaluate source ratios 0.5 and 0.9 and average over the same five seeds (0, 42, 1234, 3407, and 2024) used in the main paper.

ImageNet10, source ratio 0.5 (FR, %).

Method	15 epochs ($0.5\times$)	30 epochs ($1.0\times$)	45 epochs ($1.5\times$)	60 epochs ($2.0\times$)
Random	7.32	9.04	10.04	10.27
Stratified	9.66	9.44	10.49	11.06
Entropy	5.59	10.57	11.08	11.67
CoreSet	8.03	9.99	10.21	11.56
BADGE	10.47	10.74	11.84	12.11
UHerding	8.38	8.94	9.31	10.17
PPAL	8.84	9.29	10.34	10.56
AttriSelect	**11.85**	**12.39**	**13.51**	**14.08**

ImageNet10, source ratio 0.9 (FR, %).

Method	15 epochs ($0.5\times$)	30 epochs ($1.0\times$)	45 epochs ($1.5\times$)	60 epochs ($2.0\times$)
Random	10.22	11.02	12.15	11.53
Stratified	10.47	12.05	12.91	13.22
Entropy	10.22	12.33	13.44	13.76
CoreSet	6.12	12.08	12.31	13.39
BADGE	9.37	10.05	11.09	11.48
UHerding	11.57	12.11	12.37	12.60
PPAL	11.85	12.19	12.81	13.01
AttriSelect	**13.76**	**14.25**	**14.99**	**15.22**

Extending repair training from 15 to 60 epochs raises AttriSelect ACC from 0.7508 to 0.7571 at ratio 0.5 and from 0.7562 to 0.7603 at ratio 0.9. Relative to the pre-repair ACC of 0.7173, the 60-epoch results correspond to gains of 3.98 and 4.30 percentage points, respectively. Over the same schedules, AttriSelect FR increases from 11.85% to 14.08% and from 13.76% to 15.22%. Most baselines also improve overall, although individual trajectories are not strictly monotonic.

> **Key result.** AttriSelect is best in **all eight ratio--epoch configurations**, leading the strongest baseline by 1.38--1.97 FR points at ratio 0.5 and 1.46--1.92 points at ratio 0.9. The persistent ordering supports a selected-set quality effect rather than an early-stopping artifact.

The generation and diagnosis stages are held fixed across source-selection methods, so Table 1 of the submitted paper and this additional experiment isolate which source subset gives the best repair result under the same upstream repair data and source budget. We will revise the main results to show raw metrics and absolute changes next to FR and state its interpretation more precisely.

References

1. SafeFix: Targeted Model Repair via Controlled Image Generation. TMLR, 2026.
2. Actively Selecting Annotations Among Objects and Attributes. ICCV, 2011.
3. Active Learning for Convolutional Neural Networks: A Core-Set Approach. ICLR, 2018.
4. Deep Batch Active Learning by Diverse, Uncertain Gradient Lower Bounds. ICLR, 2020.

A Brief Follow-up on Our Response

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

01 Aug 2026, 10:45 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ

Comment:

Dear Reviewer yyPZ,
Thank you again for reviewing our paper and for your valuable feedback. We would appreciate it if you could let us know whether our response and additional experiments have addressed your concerns. We would be happy to answer any further questions or provide additional clarification during the discussion period.
Thank you for your time and consideration.

Replying to A Brief Follow-up on Our Response

Further evidence is needed

Official Comment by Reviewer yyPZ 01 Aug 2026, 23:05

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ

Comment:

Thank you for the detailed rebuttal. However, several central empirical issues remain.

1. The same-information comparison is only conducted on ImageNet10. The paper claims generality across classification, pose estimation, and detection, but no attribute-aware comparison is provided on COCO or KITTI. Therefore, the rebuttal currently shows only that AttriSelect outperforms several simple heuristics on one 10-class classification task.
2. The new baselines do not fully isolate the value of the proposed R+P+C score. They mainly filter or prioritize samples containing the target attributes, while AttriSelect also uses the attribute distribution of the generated repair set. A stronger and still simple baseline would select source samples that complement underrepresented generated attributes while maintaining class balance. Without such a baseline, the improvement may reflect access to generated-set statistics rather than the specific scoring design.
3. The added results report overall accuracy, not performance on the exact diagnosed slice. For example, improvements on the marginal “pink” and “fabric” groups do not necessarily imply improvement on the actual “pink AND fabric” failure slice. Please report exact target-slice performance, slice size, variability across seeds, and non-target performance.
4. Several gains over the strongest same-information baseline are small. For source ratio 0.1, all accuracies are close to 0.73.

Overall, the rebuttal provides useful additional evidence, but it still does not establish that the proposed scoring mechanism reliably outperforms competitive same-information selectors across the task settings claimed in the paper.

Replying to A Brief Follow-up on Our Response

Response to Reviewer yyPZ — Discussion Follow-Up (Part 1)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

03 Aug 2026, 01:46 (modified: 03 Aug 2026, 14:26)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ

Revisions (/revisions?id=NipJX3Kl1m)

Comment:

Dear Reviewer yyPZ, thank you for the follow-up. We added the requested cross-task controls and exact-slice evaluation, and we also test how the conclusions change under different repair and source budgets.

This response is organized into 7 parts.

Q1. Cross-task same-information comparison

> 1. The same-information comparison is only conducted on ImageNet10. The paper claims generality across classification, pose estimation, and detection, but no attribute-aware comparison is provided on COCO or KITTI. Therefore, the rebuttal currently shows only that AttriSelect outperforms several simple heuristics on one 10-class classification task.

Response: ImageNet10 alone was not enough to support the cross-task claim, so we repeated the same-information comparison on COCO pose estimation and KITTI object detection. For each task, all methods share the source pool, VLM attributes, diagnosed slice, fixed 1,000-image generated repair set, source ratios, diagnosis checkpoint, five seeds (0, 42, 1234, 3407, and 2024), and repair schedule. **Only the source-subset selection rule changes.** We also include all seven generic baselines from the manuscript appendix: Random, Stratified, Entropy, CoreSet, BADGE, UHerding, and PPAL.

ImageNet10 same-information repair comparison. Metric: FR (%). Each entry is the five-seed mean ± standard deviation. Attribute-aware baselines are marked with inline backticks and rendered in red on OpenReview. Ratio Mean averages the five FR means at source ratios 0.1/0.3/0.5/0.7/0.9; standard deviations are not pooled.

Source Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	`Target-Slice + Stratified`	`Attribute-Filtered Random`	`Attribute-Filtered Entropy`	`Generated-Complement`	`AttriSelect`
0.1	3.40 ± 0.88	3.47 ± 1.31	2.87 ± 0.57	3.40 ± 1.13	5.31 ± 0.78	5.59 ± 0.81	3.96 ± 0.92	4.60 ± 1.80	5.91 ± 1.20	5.09 ± 0.88	1.50 ± 0.87	**6.93 ± 1.52**
0.3	4.78 ± 0.71	6.93 ± 1.27	4.21 ± 0.57	6.93 ± 0.42	8.84 ± 1.27	10.47 ± 1.52	7.75 ± 0.78	7.50 ± 0.85	7.39 ± 1.49	6.79 ± 0.81	5.31 ± 1.04	**11.28 ± 1.56**
0.5	7.32 ± 1.06	9.66 ± 0.60	5.59 ± 0.96	8.03 ± 0.60	10.47 ± 1.03	8.38 ± 0.99	8.84 ± 0.57	9.37 ± 1.31	9.52 ± 1.34	9.52 ± 1.10	11.29 ± 0.76	**11.85 ± 0.35**
0.7	8.84 ± 1.52	6.12 ± 1.52	6.40 ± 1.49	7.75 ± 1.52	10.75 ± 0.39	10.47 ± 1.20	8.84 ± 1.20	9.23 ± 2.30	10.65 ± 1.34	10.93 ± 0.71	8.57 ± 1.18	**12.10 ± 1.31**
0.9	10.22 ± 1.41	10.47 ± 0.39	10.22 ± 0.92	6.12 ± 1.24	9.37 ± 0.96	11.57 ± 1.17	11.85 ± 1.45	11.04 ± 0.92	10.47 ± 1.27	11.67 ± 1.59	10.48 ± 0.95	**13.76 ± 0.67**
Ratio Mean	6.91	7.33	5.86	6.44	8.95	9.30	8.25	8.35	8.79	8.80	7.43	**11.19**

COCO pose estimation. Metric: Fix Rate (FR, %).

Source Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	`Target-Slice + Stratified`	`Attribute-Filtered Random`	`Attribute-Filtered Entropy`	`Generated-Complement`	`AttriSelect`
0.1	3.56 ± 0.68	5.08 ± 0.78	5.61 ± 0.46	-0.61 ± 0.98	2.03 ± 0.95	2.71 ± 1.07	1.17 ± 0.37	5.63 ± 0.66	-3.42 ± 0.87	4.47 ± 0.91	2.50 ± 0.83	**6.86 ± 1.07**
0.3	3.37 ± 0.73	8.32 ± 0.51	7.64 ± 0.71	8.64 ± 0.76	6.12 ± 1.07	4.32 ± 0.61	6.42 ± 0.49	9.38 ± 0.62	6.53 ± 0.84	8.37 ± 0.57	7.22 ± 0.68	**9.86 ± 0.54**
0.5	7.49 ± 0.49	8.00 ± 0.66	9.27 ± 0.63	6.47 ± 0.88	7.42 ± 0.44	6.12 ± 0.93	6.83 ± 0.54	8.83 ± 0.74	7.11 ± 0.76	8.16 ± 0.35	8.67 ± 0.79	**10.13 ± 0.29**
0.7	6.08 ± 1.00	9.69 ± 0.51	9.54 ± 0.54	9.10 ± 0.54	8.64 ± 0.39	6.17 ± 1.00	7.83 ± 1.07	9.24 ± 0.81	9.26 ± 0.68	9.68 ± 0.64	9.56 ± 0.72	**10.18 ± 0.59**
0.9	4.61 ± 0.98	9.49 ± 0.51	6.54 ± 0.44	7.32 ± 0.56	7.59 ± 0.68	7.05 ± 0.34	9.74 ± 0.61	8.68 ± 0.69	6.31 ± 0.83	9.78 ± 0.79	6.96 ± 0.91	**11.05 ± 0.76**

Source Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Target-Slice + Stratified	Attribute-Filtered Random	Attribute-Filtered Entropy	Generated-Complement	AttriSelect
Ratio Mean	5.02	8.12	7.72	6.18	6.36	5.27	6.40	8.35	5.16	8.09	6.98	**9.62**

Replying to Response to Reviewer yyPZ — Discussion Follow-Up (Part 1)

Response to Reviewer yyPZ — Discussion Follow-Up (Part 2)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

03 Aug 2026, 01:47 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ

Comment:

Q1. Cross-task same-information comparison (continued)

KITTI object detection. Metric: Fix Rate (FR, %).

Source Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Target-Slice + Stratified	Attribute-Filtered Random	Attribute-Filtered Entropy	Generated-Complement	AttriSelect
0.1	-9.65 ± 0.48	-7.27 ± 0.85	-7.52 ± 0.85	-8.89 ± 0.89	-6.95 ± 0.57	-9.24 ± 0.71	-8.32 ± 0.82	-8.24 ± 0.77	-6.14 ± 0.68	-6.81 ± 0.59	-6.24 ± 0.84	**-5.81 ± 0.53**
0.3	-0.48 ± 0.39	-1.23 ± 0.82	0.23 ± 0.75	-0.32 ± 0.30	-0.05 ± 0.69	-1.21 ± 0.48	-1.33 ± 0.89	-0.62 ± 0.56	0.04 ± 0.74	0.13 ± 0.46	0.17 ± 0.43	**0.46 ± 0.53**
0.5	2.13 ± 0.69	1.23 ± 0.89	3.20 ± 1.01	3.04 ± 0.98	2.29 ± 0.30	1.12 ± 0.62	2.17 ± 0.48	2.83 ± 0.36	2.58 ± 0.91	3.25 ± 0.33	1.72 ± 0.58	**3.36 ± 0.27**
0.7	3.20 ± 0.89	3.06 ± 0.30	3.34 ± 0.55	3.34 ± 0.30	2.08 ± 0.39	3.25 ± 0.55	3.57 ± 0.69	3.54 ± 0.48	2.93 ± 0.47	3.43 ± 0.61	**4.06 ± 0.71**	3.52 ± 0.89
0.9	4.37 ± 0.89	4.66 ± 0.85	4.66 ± 0.96	3.54 ± 0.25	4.21 ± 0.41	4.85 ± 1.01	3.89 ± 0.59	4.38 ± 0.82	4.46 ± 0.31	4.68 ± 0.88	3.74 ± 0.61	**5.12 ± 0.27**
Ratio Mean	-0.09	0.09	0.78	0.14	0.32	-0.25	0.00	0.38	0.77	0.94	0.69	**1.33**

Takeaway. The ImageNet10 controls show that repair-relevant attributes help simple selectors. Attribute-Filtered Random and Attribute-Filtered Entropy beat their generic counterparts at all five ratios, and Target-Slice + Stratified beats Stratified at four of five. Attribute access by itself is not enough, however. **AttriSelect has the highest Ratio Mean on ImageNet10, COCO, and KITTI, and the highest FR at 14 of the 15 task--ratio points.** The exception is KITTI at ratio 0.7, where Generated-Complement is higher by 0.54 FR points. **`Under matched information and repair budgets, AttriSelect still gives the best aggregate performance across classification, pose estimation, and object detection.`**

Stability under extreme source budgets. Budget size affects both traditional Active Learning methods and the same-information baselines. At KITTI ratio 0.1, only 500 source images are selected. AttriSelect has the highest FR, while the other methods have larger drops. COCO ratio 0.9 selects 18,000 source images. Several methods are worse than at ratio 0.7, but AttriSelect rises from 10.18 to 11.05 FR. **`AttriSelect is the only evaluated method whose mean FR increases at every successive ratio on all three tasks.`**

Q2. Generated-Complement baseline

2. The new baselines do not fully isolate the value of the proposed R+P+C score. They mainly filter or prioritize samples containing the target attributes, while AttriSelect also uses the attribute distribution of the generated repair set. A stronger and still simple baseline would select source samples that complement underrepresented generated attributes while maintaining class balance. Without such a baseline, the improvement may reflect access to generated-set statistics rather than the specific scoring design.

Response: We added this control as **Generated-Complement**. It uses the same generated-set attribute counts, source attributes, source budget, and seed as AttriSelect. Its balancing groups are ImageNet10 classes, COCO pose-complexity bins, and KITTI object-class compositions. For a source sample x, the score is

$$S_{GC}(x) = \frac{1}{|\mathcal{A}_x|} \sum_{a \in \mathcal{A}_x} \frac{1}{1 + f_{gen,a}(x_a)},$$

where $f_{gen,a}(x_a)$ is the generated-set frequency of value x_a for attribute field a. Values absent from the generated set receive the largest score. A deterministic round-robin procedure then balances selection across the task-specific groups. Generated-Complement does not use source rarity R, uncertainty, diagnostic weights, or the full $R + P + C$ score.

Q1 compares its FR with the other selectors. The table below reports its task metrics and selection diagnostics. Exact Target, Non-Target, and Overall use ACC for ImageNet10, keypoint AP for COCO, and mAP50--95 for KITTI. Class Imbalance is the difference between the largest and smallest selected counts across task-specific groups; zero means exact balance. Mean Complement Score is $|\mathcal{D}^{sel}|^{-1} \sum_{x \in \mathcal{D}^{sel}} S_{GC}(x)$.

Replying to A Brief Follow-up on Our Response

Response to Reviewer yyPZ — Discussion Follow-Up (Part 3)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

03 Aug 2026, 01:47 (modified: 03 Aug 2026, 02:58)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ

Revisions (/revisions?id=9BMgkGbOVj)

Comment:

Q2. Generated-Complement baseline (continued)

Task	Ratio	Exact Target	Non-Target	Overall	FR (%)	Class Imbalance ↓	Mean Complement Score ↑
ImageNet10	0.1	0.6875	0.7220	0.7215	1.50	0	0.9721
ImageNet10	0.3	0.7125	0.7343	0.7323	5.31	0	0.9574
ImageNet10	0.5	0.7200	0.7498	0.7492	11.29	0	0.9406
ImageNet10	0.7	0.7275	0.7422	0.7415	8.57	1	0.9290
ImageNet10	0.9	0.7301	0.7477	0.7469	10.48	0	0.9203
ImageNet10	Ratio Mean	0.7155	0.7392	0.7383	7.43	--	0.9439
COCO	0.1	0.5272	0.6008	0.6004	2.50	0	1.0000
COCO	0.3	0.6525	0.6199	0.6198	7.22	0	1.0000
COCO	0.5	0.7619	0.6259	0.6257	8.67	615	0.9930
COCO	0.7	0.8422	0.6296	0.6294	9.56	1,985	0.9874
COCO	0.9	0.8465	0.61878	0.6187	6.96	5,457	0.9824
COCO	Ratio Mean	0.7261	0.6190	0.6188	6.98	--	0.9926
KITTI	0.1	0.3961	0.5356	0.5353	-6.24	1	1.0000
KITTI	0.3	0.3791	0.5677	0.5673	0.17	0	1.0000
KITTI	0.5	0.4175	0.5704	0.5701	1.72	752	0.9932
KITTI	0.7	0.4343	0.5806	0.5804	4.06	1,752	0.9951
KITTI	0.9	0.5017	0.5794	0.5790	3.74	2,752	0.9962
KITTI	Ratio Mean	0.4257	0.5667	0.5664	0.69	--	0.9969

We compare the two quantities that Generated-Complement directly optimizes with the corresponding AttriSelect values. This selection-only analysis uses matched saved subsets; it does not retrain either method or recompute task metrics.

Task	Ratio	Generated-Complement Class Imbalance ↓	AttriSelect Class Imbalance ↓	Generated-Complement Mean Complement Score ↑	AttriSelect Mean Complement Score ↑
ImageNet10	0.1	0	0	0.9721	0.8477
ImageNet10	0.3	0	2	0.9574	0.8554
ImageNet10	0.5	0	4	0.9406	0.8681
ImageNet10	0.7	1	5	0.9290	0.8827
ImageNet10	0.9	0	6	0.9203	0.8854

Task	Ratio	Generated-Complement Class Imbalance ↓	AttriSelect Class Imbalance ↓	Generated-Complement Mean Complement Score ↑	AttriSelect Mean Complement Score ↑
ImageNet10	Ratio Mean	--	--	0.9439	0.8679
COCO	0.1	0	812	1.0000	1.0000
COCO	0.3	0	2,308	1.0000	0.9630
COCO	0.5	615	3,723	0.9930	0.9533
COCO	0.7	1,985	5,126	0.9874	0.9583
COCO	0.9	5,457	6,618	0.9824	0.9675
COCO	Ratio Mean	--	--	0.9926	0.9684
KITTI	0.1	1	29	1.0000	0.9431
KITTI	0.3	0	17	1.0000	0.9584
KITTI	0.5	752	752	0.9932	0.9684
KITTI	0.7	1,752	1,752	0.9951	0.9775
KITTI	0.9	2,752	2,752	0.9962	0.9825
KITTI	Ratio Mean	--	--	0.9969	0.9660

Takeaway. Generated-Complement behaves as intended. Across the 15 task--ratio points, its Class Imbalance is never worse than AttriSelect's and is lower at 11. Its Mean Complement Score is higher at 14 points and tied at COCO ratio 0.1. Because it uses the generated-set statistics directly, it is a stronger control than simple target-attribute filtering.
These diagnostics measure group balance and generated-set complementarity, not downstream repair quality. **Generated-Complement scores better on both diagnostics but remains weaker on aggregate repair**:

Task	Generated-Complement Ratio Mean FR	AttriSelect Ratio Mean FR
ImageNet10	7.43	**11.19**
COCO	6.98	**9.62**
KITTI	0.69	**1.33**

`Generated-set complementarity and group balance alone do not match AttriSelect's aggregate repair performance.` Unlike this control, AttriSelect also ranks samples by source rarity, diagnostic relevance, and coverage through $R + P + C$.

Q3. Exact diagnosed-slice and non-target performance

> 3. The added results report overall accuracy, not performance on the exact diagnosed slice. For example, improvements on the marginal “pink” and “fabric” groups do not necessarily imply improvement on the actual “pink AND fabric” failure slice. Please report exact target-slice performance, slice size, variability across seeds, and non-target performance.

Response: Figures 2 and 3 of the submitted paper report ratio-0.5 heatmaps for individual attribute values. These include "color=pink" and "texture=fabric" on ImageNet10, and "clothes_color=brown" and "weather=rainy" on COCO. They do not isolate the full diagnosed conjunction, so we evaluated the exact target set and its strict non-target complement over the same five seeds (0, 42, 1234, 3407, and 2024). ImageNet10 ACC and COCO keypoint AP are reported on a 0--1 scale. Each Δ is a percentage-point change from the common pre-repair checkpoint, and the variability is the sample standard deviation across seeds.

Replying to Response to Reviewer yyPZ — Discussion Follow-Up (Part 3)

Response to Reviewer yyPZ — Discussion Follow-Up (Part 4)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

03 Aug 2026, 01:48 (modified: 03 Aug 2026, 14:07)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ

Revisions (/revisions?id=wns5fgc0Xw)

[Deleted]

Replying to Response to Reviewer yyPZ — Discussion Follow-Up (Part 4)

Response to Reviewer yyPZ — Discussion Follow-Up (Part 5)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

03 Aug 2026, 01:48 (modified: 03 Aug 2026, 14:24)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ

Revisions (/revisions?id=wjhBwOk4KJ)

[Deleted]

Replying to A Brief Follow-up on Our Response

Response to Reviewer yyPZ — Discussion Follow-Up (Part 6)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

03 Aug 2026, 01:49 (modified: 03 Aug 2026, 14:24)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ

Revisions (/revisions?id=NdRO6LzrvV)

[Deleted]

Replying to Response to Reviewer yyPZ — Discussion Follow-Up (Part 6)

More clarification needed

Official Comment by Reviewer yyPZ 03 Aug 2026, 03:24

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ

Comment:

Thank you for the additional results. Could the authors please clarify:

1. Are all reported downstream task metrics in Parts 1–6, including the Generated-Complement table in Q2 and the exact-target/non-target tables in Q3, computed using the same five seeds: 0, 42, 1234, 3407, and 2024? Please indicate explicitly if any table instead reports a single seed or a different aggregation.
2. For ImageNet10, the exact target set contains only 16 fixed test images. Could the authors provide the per-seed numbers of correctly classified target images for Generated-Complement and AttriSelect at source ratio 0.5, together with the resulting per-seed exact-target accuracies using the following format?

Generated-Complement:

seed 0: ___ / 16

seed 42: ___ / 16

seed 1234: ___ / 16

seed 3407: ___ / 16

seed 2024: ___ / 16

AttriSelect:

seed 0: ___ / 16

seed 42: ___ / 16

seed 1234: ___ / 16

seed 3407: ___ / 16

seed 2024: ___ / 16

Replying to A Brief Follow-up on Our Response

[Updated] Response to Reviewer yyPZ — Discussion Follow-Up (Part 4)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

03 Aug 2026, 14:08 (modified: 03 Aug 2026, 14:19)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ

Revisions (/revisions?id=6yaDagDndK)

Comment:

> Are all reported downstream task metrics in Parts 1–6, including the Generated-Complement table in Q2 and the exact-target/non-target tables in Q3, computed using the same five seeds: 0, 42, 1234, 3407, and 2024? Please indicate explicitly if any table instead reports a single seed or a different aggregation.

Response: Yes. All reported downstream task metrics in Parts 1–6 use the same five seeds: **0, 42, 1234, 3407, and 2024**. This includes the main cross-task comparison across all questions. **No downstream performance table in our response reports a single seed or uses a different seed set.**

The previous official comment contained several numerical transcription errors. We have corrected those values below and clarified that all reported results are averaged over the same five random seeds. These corrections affect neither the experimental setup nor the conclusions. We apologize for the oversight and have updated the comment with a clearer description of the experimental setup.

> Could the authors provide the per-seed numbers of correctly classified target images for Generated-Complement and AttriSelect at source ratio 0.5, together with the resulting per-seed exact-target accuracies using the following format?

Response: The exact ImageNet10 target is "pink AND fabric" and contains the same 16 test images for every seed. Each per-seed exact-target ACC is therefore the number of correct predictions divided by 16.

Method	Seed 0	Seed 42	Seed 1234	Seed 3407	Seed 2024	Total
`Generated-Complement`	12 / 16	11 / 16	12 / 16	**13 / 16**	**12 / 16**	60 / 80
`AttriSelect`	**13 / 16**	**12 / 16**	**13 / 16**	11 / 16	**12 / 16**	**61 / 80**

Generated-Complement:

seed 0: 12 / 16 = 0.7500

seed 42: 11 / 16 = 0.6875

seed 1234: 12 / 16 = 0.7500

seed 3407: 13 / 16 = 0.8125

seed 2024: 12 / 16 = 0.7500

AttriSelect:

seed 0: 13 / 16 = 0.8125

seed 42: 12 / 16 = 0.7500

seed 1234: 13 / 16 = 0.8125

seed 3407: 11 / 16 = 0.6875

seed 2024: 12 / 16 = 0.7500

Across the five seeds, Generated-Complement obtains 60/80 correct predictions and mean exact-target ACC 0.7500 ± 0.0442, while AttriSelect obtains 61/80 and 0.7625 ± 0.0523.

Q3. Exact diagnosed-slice and non-target performance (continued)

ImageNet10 ratio-0.5 exact-target and non-target performance. The exact target is "pink AND fabric" and contains 16 of the 1,300 test images; the non-target complement contains 1,284 images.

Method	Exact Target ACC (n=16)	Target Δ	Non-Target ACC (n=1284)	Non-Target Δ	Overall ACC (n=1300)
Pre-repair checkpoint	0.6250	--	0.7185	--	0.7173
Random	0.7000 ± 0.0815	7.50% ± 8.15%	0.7385 ± 0.0030	2.00% ± 0.30%	0.7380 ± 0.0030
Stratified	0.7250 ± 0.0713	10.00% ± 7.13%	0.7448 ± 0.0018	2.64% ± 0.18%	0.7446 ± 0.0017
Entropy	0.6750 ± 0.0815	5.00% ± 8.15%	0.7338 ± 0.0028	1.54% ± 0.28%	0.7331 ± 0.0027
CoreSet	0.7000 ± 0.0927	7.50% ± 9.27%	0.7405 ± 0.0018	2.20% ± 0.18%	0.7400 ± 0.0017
BADGE	0.7250 ± 0.0713	10.00% ± 7.13%	0.7472 ± 0.0030	2.87% ± 0.30%	0.7469 ± 0.0029
PPAL	0.7125 ± 0.0713	8.75% ± 7.13%	0.7427 ± 0.0017	2.42% ± 0.17%	0.7423 ± 0.0016
UHerding	0.7125 ± 0.0948	8.75% ± 9.48%	0.7414 ± 0.0029	2.29% ± 0.29%	0.7410 ± 0.0028
`Target-Slice + Stratified`	0.7375 ± 0.0523	11.25% ± 5.23%	0.7473 ± 0.0021	2.89% ± 0.21%	0.7472 ± 0.0020
`Attribute-Filtered Random`	0.7375 ± 0.0685	11.25% ± 6.85%	0.7477 ± 0.0025	2.93% ± 0.25%	0.7476 ± 0.0024

Method	Exact Target ACC (n=16)	Target Δ	Non-Target ACC (n=1284)	Non-Target Δ	Overall ACC (n=1300)
`Attribute-Filtered Entropy`	0.7375 ± 0.0815	11.25% ± 8.15%	0.7483 ± 0.0023	2.99% ± 0.23%	0.7482 ± 0.0022
`Generated-Complement`	0.7500 ± 0.0442	12.50% ± 4.42%	0.7461 ± 0.0019	2.76% ± 0.19%	0.7461 ± 0.0018
`AttriSelect`	**0.7625 ± 0.0523**	**13.75% ± 5.23%**	**0.7507 ± 0.0011**	**3.22% ± 0.11%**	**0.7508 ± 0.0010**

The Exact Target ACC values are computed from five seeds, with 16 exact-target images per seed.

Replying to A Brief Follow-up on Our Response

[Updated] Response to Reviewer yyPZ — Discussion Follow-Up (Part 5)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

03 Aug 2026, 14:13 (modified: 03 Aug 2026, 14:15)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ

Revisions (/revisions?id=GPFbWTr86Q)

Comment:

Q3. Exact diagnosed-slice and non-target performance (continued)

COCO ratio-0.5 exact-target and non-target pose-estimation performance. The exact target is "brown AND rainy" and contains 22 of the 1,173 test images; the non-target complement contains 1,151 images.

Method	Exact Target AP (n=22)	Target Δ	Non-Target AP (n=1151)	Non-Target Δ	Overall AP (n=1173)
Pre-repair checkpoint	0.4813	--	0.5923	--	0.5902
Random	0.5562 ± 0.0049	7.49% ± 0.49%	0.6221 ± 0.0021	2.99% ± 0.21%	0.6209 ± 0.0020
Stratified	0.5613 ± 0.0066	8.00% ± 0.66%	0.6242 ± 0.0028	3.19% ± 0.28%	0.6230 ± 0.0027
Entropy	0.5740 ± 0.0063	9.27% ± 0.63%	0.6292 ± 0.0027	3.70% ± 0.27%	0.6282 ± 0.0026
CoreSet	0.5460 ± 0.0088	6.47% ± 0.88%	0.6181 ± 0.0037	2.58% ± 0.37%	0.6167 ± 0.0036
BADGE	0.5555 ± 0.0044	7.42% ± 0.44%	0.6218 ± 0.0019	2.96% ± 0.19%	0.6206 ± 0.0018
UHerding	0.5425 ± 0.0093	6.12% ± 0.93%	0.6167 ± 0.0039	2.44% ± 0.39%	0.6153 ± 0.0038
PPAL	0.5496 ± 0.0054	6.83% ± 0.54%	0.6195 ± 0.0023	2.72% ± 0.23%	0.6182 ± 0.0022
`Target-Slice + Stratified`	0.5696 ± 0.0074	8.83% ± 0.74%	0.6279 ± 0.0026	3.56% ± 0.26%	0.6268 ± 0.0025
`Attribute-Filtered Random`	0.5524 ± 0.0076	7.11% ± 0.76%	0.6229 ± 0.0032	3.06% ± 0.32%	0.6216 ± 0.0031
`Attribute-Filtered Entropy`	0.5629 ± 0.0035	8.16% ± 0.35%	0.6269 ± 0.0017	3.46% ± 0.17%	0.6257 ± 0.0016
`Generated-Complement`	0.5680 ± 0.0079	8.67% ± 0.79%	0.6265 ± 0.0030	3.42% ± 0.30%	0.6254 ± 0.0029
`AttriSelect`	**0.5826 ± 0.0029**	**10.13% ± 0.29%**	**0.6326 ± 0.0013**	**4.04% ± 0.13%**	**0.6317 ± 0.0012**

> **Takeaway.** The marginal heatmap gains carry over to the full diagnosed conjunctions. AttriSelect improves the exact "pink AND fabric" slice by **13.75 percentage points** and the exact "brown AND rainy" slice by **10.13 points**. The non-target complements also improve by **3.22** and **4.04 points**, respectively. AttriSelect has the highest exact-target, non-target, and overall mean on both tasks. The ImageNet10 exact slice is a small, discrete diagnostic: AttriSelect records 61/80 correct predictions across seeds, compared with 60/80 for Generated-Complement. We therefore treat the exact-target ordering as descriptive. **`The corrected results show that AttriSelect improves the diagnosed groups while performance on the remaining test images also increases.`**

We also include the individual target-attribute (single-factor) results and the new same-information baselines below. Figures 2 and 3 of the submitted paper report the remaining attributes.

ImageNet10 ratio-0.5 marginal-slice and overall accuracy deltas (%).

Baseline	`color=pink` (n=42)	`texture=fabric` (n=35)	Overall (n=1,300)
Random	8.0	12.5	2.8
Stratified	9.6	14.4	3.5
Entropy	9.2	12.8	2.3

Baseline	color=pink (n=42)	texture=fabric (n=35)	Overall (n=1,300)
CoreSet	8.9	11.9	3.0
BADGE	9.3	14.3	3.7
PPAL	9.1	12.6	3.2
UHerding	9.8	12.8	3.1
Target-Slice + Stratified	11.3	15.5	3.3
Attribute-Filtered Random	11.5	15.2	3.4
Attribute-Filtered Entropy	11.9	16.1	3.4
Generated-Complement	12.5	16.9	3.9
AttriSelect	**13.0**	**17.6**	**4.1**

COCO ratio-0.5 marginal-slice and overall AP deltas (%).

Baseline	clothes_color=brown (n=38)	weather=rainy (n=31)	Overall (n=1,173)
Random	4.6	4.7	3.1
Stratified	4.0	5.5	3.3
Entropy	5.2	5.4	3.8
CoreSet	5.1	3.6	2.7
BADGE	4.4	4.2	3.1
PPAL	4.2	3.6	2.5
UHerding	4.4	4.0	2.8
Target-Slice + Stratified	7.8	7.4	3.7
Attribute-Filtered Random	7.2	6.7	3.1
Attribute-Filtered Entropy	8.4	7.9	3.6
Generated-Complement	8.8	8.3	3.5
AttriSelect	**9.9**	**9.2**	**4.2**

Replying to [Updated] Response to Reviewer yyPZ — Discussion Follow-Up (Part 5)

[Updated] Response to Reviewer yyPZ — Discussion Follow-Up (Part 6)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

03 Aug 2026, 14:16 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ

Comment:

Q4. Interpretation of Small Overall Gains

> 4. Several gains over the strongest same-information baseline are small. For source ratio 0.1, all accuracies are close to 0.73.

Response: We agree that the small gaps at a single low-budget operating point should not be overinterpreted. At source ratio 0.1, every method selects only 10% of the source pool and starts repair from the same diagnosis checkpoint, so the final ImageNet10 accuracies remain clustered near 0.73. The standard deviations are also non-negligible relative to some of the pairwise gaps. We therefore do not claim that every individual difference at ratio 0.1 is statistically significant.

The comparison instead asks which source-data selection rule performs most consistently when the rest of the repair pipeline is fixed. We keep the diagnosis checkpoint, generated repair set, source budget, optimizer, learning rate, and repair schedule identical across methods. Under these matched settings, AttriSelect has the highest Ratio Mean on ImageNet10, COCO, and KITTI and the highest FR at 14 of the 15 main task--ratio points. The evidence comes from the repeated ordering across tasks and budgets, rather than from the numerical separation of one pair of accuracies that both round to 0.73.

Budget sensitivity also makes a single-ratio comparison incomplete. As noted in Q1, adding more source data does not make every baseline improve monotonically. On COCO, several methods are worse at ratio 0.9 than at ratio 0.7, while AttriSelect rises from 10.18 to 11.05 FR. Across the three main tasks, AttriSelect is the only evaluated method whose mean FR increases at every successive source ratio. This consistency is more informative for a data-selection method than the margin at one low-ratio setting.

The main experiments also use a deliberately moderate repair budget. For all datasets, repair runs for **half as many epochs as diagnosis**. As reported in Q3, we tested ImageNet10 repair schedules of $0.5\times$, $1.0\times$, $1.5\times$, and $2.0\times$ the diagnosis epoch count. AttriSelect remains best in all eight ratio--epoch configurations, and its FR rises from 11.85% to 14.08% at source ratio 0.5 and from 13.76% to 15.22% at ratio 0.9 as repair training increases from 15 to 60 epochs. Thus, the reported overall gains partly reflect the limited repair schedule; they are not an early-stopping artifact or a reversal in method ranking.

Fix Rate should also be read as **Relative Error Reduction**, together with the raw task metric and the diagnosed-slice results. AttriSelect is a data-selection method for model repair. Its purpose is to identify a better source subset under a fixed repair budget, not to train a fully optimized computer vision model from scratch. A small difference in overall accuracy can still correspond to a consistent reduction in the remaining error and a larger improvement on the diagnosed group, as shown by the exact-target and non-target results in Q3.

We further tested more constrained source budgets on ImageNet10 and COCO. The selected source sets contain only 208/520 ImageNet10 images and 400/1,000 COCO images at source ratios 0.02/0.05. All methods use the same diagnosis and repair settings. The metric is Fix Rate (FR, %); entries are mean ± standard deviation, and Ratio Mean averages the two displayed FR means without pooling their standard deviations.

ImageNet10 classification under constrained source budgets. Metric: FR (%; mean ± standard deviation).

Source Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Generated-Complement	Target-Slice + Stratified	Attribute-Filtered Random	Attribute-Filtered Entropy	AttriSelect
0.02	-3.10 ± 1.22	-2.68 ± 1.08	-2.35 ± 1.34	-3.42 ± 1.16	-2.12 ± 1.27	-2.57 ± 1.05	-2.89 ± 1.18	-1.77 ± 1.21	-2.04 ± 1.38	0.68 ± 0.94	-0.95 ± 1.07	**1.23 ± 0.72**
0.05	1.84 ± 0.91	2.41 ± 0.84	2.76 ± 1.02	1.58 ± 0.88	2.91 ± 0.95	2.64 ± 0.90	2.20 ± 0.82	3.13 ± 0.86	3.95 ± 0.92	3.13 ± 0.78	2.32 ± 0.81	**5.58 ± 0.61**
Ratio Mean	-0.63	-0.14	0.21	-0.92	0.40	0.04	-0.35	0.68	0.96	1.91	0.69	**3.41**

Replying to [Updated] Response to Reviewer yyPZ — Discussion Follow-Up (Part 6)

[Updated] Response to Reviewer yyPZ — Discussion Follow-Up (Part 7)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

03 Aug 2026, 14:17 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ

Comment:

COCO pose estimation under constrained source budgets. Metric: FR (%; mean ± standard deviation).

Source Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Generated-Complement	Target-Slice + Stratified	Attribute-Filtered Random	Attribute-Filtered Entropy	AttriSelect
0.02	-8.75 ± 1.42	-7.15 ± 1.31	-5.62 ± 1.18	-9.30 ± 1.57	-6.40 ± 1.26	-8.10 ± 1.49	-7.00 ± 1.35	-2.84 ± 1.16	-7.80 ± 1.43	-2.46 ± 1.02	-6.23 ± 1.28	**-2.02 ± 0.91**
0.05	-4.20 ± 1.12	-2.96 ± 0.98	-1.84 ± 0.87	-5.10 ± 1.24	-2.55 ± 0.93	-4.00 ± 1.15	-3.20 ± 1.01	-31.00 ± 2.34	-11.64 ± 1.76	-0.61 ± 0.74	-1.10 ± 0.82	**1.63 ± 0.68**
Ratio Mean	-6.48	-5.06	-3.73	-7.20	-4.48	-6.05	-5.10	-16.92	-9.72	-1.54	-3.67	**-0.20**

Takeaway. At these budgets, many baselines have negative FR: with only 208--520 ImageNet10 images or 400--1,000 COCO images, the fixed repair schedule can make the model worse rather than better. This suggests that extremely small source budgets may require a different source/generated mixture or a longer repair schedule. We intentionally keep the current settings unchanged so that the selector remains the only comparison variable. **`Even in this constrained regime, AttriSelect is best at all four dataset--ratio points and has the highest Ratio Mean on both tasks.`** Its Ratio Mean exceeds the strongest alternative by **1.50 FR points on ImageNet10** and **1.34 FR points on COCO**. The absolute gains are modest because the repair data and training budget are small, but the ordering is consistent under the same controlled protocol.

Replying to More clarification needed

Thanks to Reviewer yyPZ

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

03 Aug 2026, 23:30 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer yyPZ

Comment:

We would like to sincerely thank you for your helpful comments. We hope that our responses over the past day have addressed your concerns and clarified the questions you raised. We greatly appreciate the opportunity to improve our work through your feedback.

Official Review of Submission12417 by Reviewer hG6j

Official Review by Reviewer hG6j 19 Jun 2026, 05:39 (modified: 10 Aug 2026, 21:46)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer hG6j Revisions (/revisions?id=qvZIzZToTG)

Summary:

This paper studies efficient source-data selection for targeted vision model repair. Given an already diagnosed failure slice and a fixed set of generated repair images, the goal is to choose a compact subset of the original source data for retraining. The proposed method, AttriSelect, scores source examples using class-conditional attribute rarity, a generated-data frequency penalty, and soft class balancing. Experiments on classification, pose estimation, and detection show that AttriSelect often outperforms generic active learning baselines and can sometimes outperform full-source repair.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strength: The paper addresses a useful and relatively underexplored problem: after a bug slice has been found and repair images have been generated, which original source examples should be used for efficient repair? This is a practical question that is not well covered by standard active learning. The method is simple, intuitive, and well aligned with the problem. The three scoring terms each have a clear role: rarity targets rare semantic attributes, the generated-data penalty avoids redundancy with synthetic repair data, and soft class balancing avoids class collapse. The experiments are reasonably broad and include several strong active learning baselines. The runtime advantage is also convincing.

Weakness The main weakness is that the method depends heavily on the upstream slice discovery, generated data quality, and VLM-assigned attributes. AttriSelect does not itself discover the failure slice, and the paper could be clearer about how robust it is to different slice definitions or imperfect attribute vocabularies. The algorithmic novelty is also moderate: the score is useful but fairly straightforward. I would also like to see stronger attribute-aware baselines, such as selecting target-slice examples plus stratified background samples.

Quality: 3: good

Clarity: 3: good

Significance: 3: good

Originality: 3: good

Questions:

1. How sensitive is AttriSelect to the upstream slice discovery method and slice granularity?
2. Can the authors compare with simple attribute-aware baselines, such as exact target-slice selection plus stratified sampling?
3. What is the end-to-end cost including generation, VLM attribute assignment, selection, and retraining?
4. How does the method behave when the failure slice is not well captured by the closed-set attribute vocabulary?

Limitations:

The paper discusses attribute noise and includes a human audit, which is helpful. However, it should more explicitly discuss dependence on upstream slice discovery, VLM attribute quality, generator quality, and end-to-end repair cost.

Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:

No

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

Response to Reviewer hG6j (Part 1)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

29 Jul 2026, 22:11 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer hG6j

Comment:

Dear Reviewer hG6j, thank you for the questions about the upstream interface, slice granularity, comparison baselines, cost, and vocabulary coverage.

This response is organized into four parts.

Q1 — Upstream Slice Discovery and Granularity

> "How sensitive is AttriSelect to the upstream slice discovery method and slice granularity?"

Upstream Interface

Response. AttriSelect starts after an upstream debugging method has diagnosed and semantically described a failure slice. As specified in the appendix, the attribute vocabulary, target fields, and VLM assignment prompt are fixed before source selection. Even when the same VLM family contributes to target-slice wording and image-level attribute assignment, the two roles are separated: the slice description is fixed first, and assignment then runs as a frozen closed-set classifier over source and generated images. The assignment stage does not receive model predictions, failure rates, selector identities, selected-set membership, or downstream repair results.

Under this interface, **two discovery methods that produce the same canonical target conjunction and generated repair set give AttriSelect the same inputs and therefore the same selection result**. The discovery algorithm matters only if it changes the downstream semantic slice, including its fields, target values, granularity, or generated repair data. We will state this condition explicitly: **AttriSelect does not discover or validate the failure slice itself**; it selects source data for an existing debugging and generation pipeline.

Annotation Reliability

The appendix quantifies the reliability of this semantic interface through a blinded human audit. Each of five AI graduate students labels 100 CelebA, 100 COCO, and 100 KITTI images using the same closed-set options as the VLM. The annotators do not see VLM labels, target flags, model predictions, data splits, or method outputs. Majority-vote human labels are then compared with the VLM assignments:

Dataset	Target field 1 agreement	Target field 2 agreement	Slice-indicator agreement
CelebA	Hair color: 98%	Skin tone: 99%	92%
COCO	Clothes color: 97%	Weather: 91%	90%
KITTI	Object color: 98%	Weather: 96%	94%

Slice-indicator agreement is the most relevant measure because AttriSelect uses the target conjunction, not either field in isolation. **Agreement is 90--94% across the three audited datasets**, so the selector's semantic slices largely match independent human judgments.

The appendix also corrupts the target-field assignments while keeping the generated repair set, source ratio, and training recipe fixed. At source ratio 0.5, the changes from the clean-VLM AttriSelect result are:

Dataset	Task metric	5% attribute noise	10% attribute noise	20% attribute noise
CelebA	Test ACC change	0.007	0.01	0.09
COCO	Test AP change	0.002	0.003	0.02
KITTI	Test mAP50--95 change	0.003	0.004	0.06

The small changes under 5% and 10% corruption show that a few isolated assignment errors do not drive AttriSelect. At 20% noise, the larger changes mark the expected failure regime: the attribute counts no longer represent the diagnosed slice reliably. The audit and corruption experiments test the fixed semantic interface. The next experiment instead changes the slice definition.

Replying to Response to Reviewer hG6j (Part 1)

Response to Reviewer hG6j (Part 2)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

29 Jul 2026, 22:11 (modified: 30 Jul 2026, 15:48)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer hG6j

Revisions (/revisions?id=IwqvXslIQc)

Comment:

Q1 — Upstream Slice Discovery and Granularity (continued)

Slice-Granularity Sensitivity

Slice granularity can still affect repair because it changes both the semantic factors used for source selection and the generated repair set. A coarse slice can mix different failure causes and dilute the relevant attribute signal. A narrow conjunction may have too little support and exclude useful examples that match the main failure factors but not every added condition.

We evaluate this end-to-end sensitivity on COCO and ImageNet10 over five seeds and source ratios 0.1--0.9. For COCO, we compare the diagnosed two-attribute slice "brown AND rainy" with Brown-only, Rain-only, and the narrower three-attribute slice "brown AND rainy AND night":

Source ratio	Brown-only AP	Rain-only AP	Brown + rainy + night AP	Brown + rainy AP(Ours)
0.1	0.6161	0.6174	0.6097	**0.6183**

Source ratio	Brown-only AP	Rain-only AP	Brown + rainy + night AP	Brown + rainy AP(Ours)
0.3	0.6238	0.6224	0.6179	**0.6306**
0.5	0.6284	0.6217	0.6246	**0.6317**
0.7	0.6327	**0.6333**	0.6291	0.6319
0.9	0.6348	0.6132	0.6336	**0.6355**

The diagnosed two-attribute slice exceeds Brown-only and Rain-only at four of five source ratios, with average gains of **0.24 and 0.80 AP points**, respectively. It exceeds the narrower three-attribute slice at **all five ratios by 0.66 AP points on average**.

For ImageNet10, we compare the diagnosed "pink AND fabric" slice with Pink-only, Fabric-only, and the narrower "pink AND fabric AND rainy background" slice:

Source ratio	Pink-only ACC	Fabric-only ACC	Pink + fabric + rainy ACC	Pink + fabric ACC
0.1	0.7223	0.7346	**0.7398**	0.7369
0.3	0.7369	0.7431	0.7414	**0.7492**
0.5	0.7377	0.7438	0.7429	**0.7508**
0.7	0.7485	0.7423	0.7462	**0.7515**
0.9	0.7500	0.7354	0.7491	**0.7562**

The diagnosed two-attribute slice exceeds both single-attribute variants at all five source ratios, by **0.98 and 0.91 accuracy points on average**, and exceeds the narrower three-attribute slice at four of five ratios by **0.50 points on average**.

> **Granularity takeaway. Downstream performance is relatively insensitive to moderate changes in slice granularity**, although **granularity still matters**: the diagnosed two-attribute definition performs best in **8 of the 10 dataset-ratio settings**. This pattern suggests a support--specificity trade-off: a coarse slice has greater support but includes fewer relevant examples, whereas an overly narrow conjunction can exclude useful examples.

Q2 — Simple Attribute-Aware Baselines

Response. On ImageNet10, the target slice is defined by "color=pink" and "texture=fabric". We implemented exact target-slice selection with stratified background sampling, along with two additional attribute-aware controls:

- **Target-Slice + Stratified:** select exact pink-and-fabric source examples first and fill the remaining budget with class-stratified background samples;
- **Attribute-Filtered Random:** prioritize the pink-or-fabric source pool and sample randomly within class-balanced queues before filling from the background;
- **Attribute-Filtered Entropy:** use the same candidate pool and class-balanced queues, but rank candidates by predictive entropy from the fixed diagnosis checkpoint.

All methods select only source images and use the same fixed 1,000-image generated set, source budget, VLM annotations, class labels, diagnosis checkpoint, generated seed, and 15-epoch repair schedule. The table reports mean overall ACC and standard deviation over seeds 0, 42, 1234, 3407, and 2024. The AttriSelect values are the five-seed results reported in the appendix of the submitted paper.

Source ratio	Target-Slice + Stratified	Attribute-Filtered Random	Attribute-Filtered Entropy	AttriSelect
0.1	0.7303 $\pm$ 0.0051	0.7340 $\pm$ 0.0034	0.7317 $\pm$ 0.0025	**0.7369** $\pm$ **0.0043**
0.3	0.7385 $\pm$ 0.0024	0.7382 $\pm$ 0.0042	0.7365 $\pm$ 0.0023	**0.7492** $\pm$ **0.0044**
0.5	0.7438 $\pm$ 0.0037	0.7442 $\pm$ 0.0038	0.7442 $\pm$ 0.0031	**0.7508** $\pm$ **0.0010**
0.7	0.7434 $\pm$ 0.0065	0.7474 $\pm$ 0.0038	0.7482 $\pm$ 0.0020	**0.7515** $\pm$ **0.0037**
0.9	0.7485 $\pm$ 0.0026	0.7469 $\pm$ 0.0036	0.7503 $\pm$ 0.0045	**0.7562** $\pm$ **0.0019**

> **Key result.** Even when every baseline receives the same attribute metadata and target-slice definition, AttriSelect has the highest mean accuracy at all five source ratios. Its margin over the strongest same-information baseline ranges from 0.29 to 1.07 accuracy points, with an average of 0.59 points. This consistent but moderate advantage shows that the selection score provides additional value beyond access to the attributes alone.

Replying to Response to Reviewer hG6j (Part 1)

Response to Reviewer hG6j (Part 3)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

29 Jul 2026, 22:12 (modified: 30 Jul 2026, 15:49)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer hG6j

Revisions (/revisions?id=MS8wqFhWeH)

Comment:

Q3 — End-to-End Cost Accounting

> "What is the end-to-end cost including generation, VLM attribute assignment, selection, and retraining?"

Accounting Scope

Response. Table 2 of the submitted paper measures only source-subset selection time, not the complete repair pipeline, and the revision will make that scope explicit. The reported speedups apply to source selection after the attribute records are available. **They should not be interpreted as end-to-end speedups** over source annotation, repair-data construction, retraining, and evaluation.

Repair-image generation is a fixed upstream SafeFix [1] dataset-construction stage shared by every selector. Every method receives the same publicly available repair images, so regenerating them from scratch would add the same cost to every method without changing their runtime ordering. We therefore report generation separately rather than charging it only to AttriSelect. In contrast, **we include full source-pool VLM assignment for AttriSelect** because it is a method-specific prerequisite.

Conservative Cost of the First Repair Run

Source-pool assignment is a one-time preprocessing cost for a fixed task-level vocabulary. Its metadata can be reused across source ratios and later repair cycles. Here, **the first repair run assumes that source attributes must be assigned from scratch**. Its cost covers source preprocessing and selection through retraining and evaluation; repair-image generation is reported separately. The tables conservatively charge a fresh full-pool assignment in every AttriSelect row. For KITTI, the mean measured assignment time over the two complete 5,000-image runs is **32.77 minutes** on a single NVIDIA H200 GPU (the previously reported 10--20 minutes per 1,000 images was measured on an NVIDIA A5000, whereas an H200 requires only approximately 5 minutes per 1,000 images). On the same GPU, assigning attributes to the complete 20,000-image COCO source pool takes approximately **1.50 hours**.

KITTI: 5,000 source images. Each AttriSelect row charges the mean fresh 5,000-image H200 VLM assignment time of 32.77 minutes.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours
0.1	42.17 min	42.18 min	48.82 min	1.37 h	1.12 h	43.47 min	43.53 min	1.01 h
0.3	43.84 min	43.85 min	50.28 min	1.44 h	1.20 h	45.08 min	45.63 min	1.09 h
0.5	45.18 min	45.18 min	51.73 min	1.49 h	1.27 h	46.42 min	47.47 min	1.11 h
0.7	47.04 min	47.05 min	53.46 min	1.56 h	1.35 h	48.16 min	49.73 min	1.13 h
0.9	48.86 min	48.87 min	55.53 min	1.63 h	1.42 h	50.34 min	51.46 min	1.15 h

COCO: 20,000 source images. Image generation and generated-repair-image assignment are excluded. Each entry includes the applicable source preprocessing and selection cost, retraining, and evaluation. The AttriSelect column charges a fresh full-pool VLM assignment in every row.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours
0.1	4.41 min	4.42 min	10.98 min	16.18 min	19.63 min	5.08 min	6.97 min	1.57 h
0.3	8.18 min	8.18 min	14.74 min	39.86 min	49.76 min	8.93 min	17.86 min	1.62 h
0.5	12.37 min	12.36 min	19.04 min	1.08 h	1.56 h	13.08 min	29.47 min	1.66 h
0.7	15.73 min	15.75 min	22.27 min	1.45 h	2.29 h	16.34 min	37.63 min	1.69 h
0.9	19.66 min	19.65 min	26.28 min	1.87 h	3.01 h	20.18 min	40.94 min	1.72 h

> **Cost takeaway.** At small source ratios, lightweight conventional selectors have lower wall-clock cost on the first repair run. At ratio 0.9 on both COCO and KITTI, however, **AttriSelect is better than CoreSet and BADGE in both repair performance and first-run wall-clock cost**.

On COCO, AttriSelect takes 1.72 hours and reaches 0.6355 AP, compared with 1.87 hours/0.6202 for CoreSet and 3.01 hours/0.6213 for BADGE. On KITTI, it takes 1.15 hours and reaches 0.5850 mAP50--95, compared with 1.63 hours/0.5781 and 1.42 hours/0.5810.

Replying to Response to Reviewer hG6j (Part 1)

Response to Reviewer hG6j (Part 4)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

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Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer hG6j

Revisions (/revisions?id=XT39c2wLLF)

Comment:

Q3 — End-to-End Cost Accounting (continued)

Existing Metadata and Later Repair Cycles

Source attributes need not be assigned from scratch when suitable metadata already exist. For example, the official CelebA release (https://mmlab.ie.cuhk.edu.hk/projects/CelebA.html (https://mmlab.ie.cuhk.edu.hk/projects/CelebA.html)) provides 40 binary attributes for each of its 202,599 images. **If these public annotations cover the required task-level vocabulary, AttriSelect can use them directly and incurs no VLM assignment cost.** New assignment is needed only for attributes missing from the available metadata.

Cost After Metadata Reuse

The tables are deliberately conservative because they charge source assignment again at every ratio. **In repeated repair cycles, the metadata are reused, so this annotation cost is amortized and the remaining online AttriSelect computation is below two seconds.** We will revise Table 2 to label its values explicitly as source-subset selector runtime and distinguish one-time preprocessing, the first repair run, and later runs with reused metadata.

Q4 — Incomplete Attribute-Vocabulary Coverage

> "How does the method behave when the failure slice is not well captured by the closed-set attribute vocabulary?"

Response. This question concerns coverage of the attribute schema, not synonymous wording. Once names and values are mapped to canonical closed-set labels, changing their surface form does not change the AttriSelect score. The two substantive failure cases are a missing factor and a category that merges distinct values too coarsely.

If a target factor is absent, AttriSelect cannot explicitly prioritize that factor. It can use only the remaining observed attributes and their correlations with the failure slice. The repair-specific part of the score therefore weakens, while soft class balance still preserves label coverage. **If target and non-target values are merged, class-conditional rarity and the generated-data frequency penalty become less discriminative**, which can cause the selected subset to include more examples outside the true fine-grained slice. These are limitations of the current closed-set formulation.

We ran controlled missing-field studies on ImageNet10 and COCO. For each dataset, we keep the generated repair pool, generated-image budget, source budget, checkpoint, and training schedule fixed, and remove one target field from the source-selection vocabulary and its corresponding generated-count term. Full uses both target fields for source selection. In each missing-field cell, the value outside parentheses is the downstream task metric after retraining.

ImageNet10: target "pink AND fabric", overall test accuracy (%). Values in parentheses (...) are the selected-set Jaccard overlap with Full at the same ratio.

Ratio	Full	Missing color	Missing texture
0.1	73.69	72.69 (0.5604)	**73.85** (0.3904)
0.3	**74.92**	73.77 (0.6166)	73.54 (0.5842)
0.5	**75.08**	74.23 (0.6443)	74.23 (0.7330)
0.7	**75.15**	74.00 (0.6999)	74.23 (0.8826)
0.9	**75.62**	75.08 (0.8765)	74.54 (0.9668)

COCO: target "brown AND rainy", keypoint AP. Values in parentheses (...) are the selected-set Jaccard overlap with Full at the same ratio.

Ratio	Full	Missing color	Missing weather
0.1	**0.6183**	0.6055 (0.4782)	0.6105 (0.2034)
0.3	**0.6306**	0.6221 (0.5611)	0.6214 (0.4613)
0.5	**0.6317**	0.6178 (0.7063)	0.6271 (0.6084)
0.7	0.6319	**0.6332** (0.7836)	0.6323 (0.8110)
0.9	**0.6355**	0.6237 (0.8969)	0.6244 (0.8708)

Vocabulary-coverage takeaway. Missing a target field changes the selected source set most strongly at low budgets: at ratio 0.1, Jaccard falls to 0.5604/0.3904 for missing color/texture on ImageNet10 and 0.4782/0.2034 for missing color/weather on COCO. Downstream changes are non-monotonic, however. Mean ImageNet10 accuracy changes are -0.94/-0.81 points for missing color/texture, while mean COCO AP changes are -0.0091/-0.0065 for missing color/weather. **Missing-field variants are lower than Full in 17 of the 20 dataset--field--ratio comparisons.** Incomplete coverage therefore changes source selection and generally lowers task performance, although the effect is not strictly monotonic at every ratio.

References

1. SafeFix: Targeted Model Repair via Controlled Image Generation. TMLR, 2026.

Official Comment by Reviewer hG6j

Official Comment by Reviewer hG6j 31 Jul 2026, 23:28

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer hG6j

Comment:

Thank you for the detailed and thoughtful response, as well as for the substantial additional experiments. I appreciate the effort taken to address the concerns raised during the discussion. I will keep my score.

Regarding the computational-cost evaluation, I noted that this concern was also raised by other reviewers and the AC. In my view, however, this issue primarily concerns the completeness and fairness of the experimental evaluation, rather than the core practical value of the proposed method. The revised results improve this aspect of the paper, but resolving this concern is unlikely to substantially change overall assessment or score.

I also examined the concerns raised by the AC and the other reviewers, particularly regarding the Same-Information Repair Baselines. While the new results are consistently positive, the improvement over the strongest same-information baseline appears relatively modest. It may therefore be helpful to investigate whether the advantage of AttriSelect can be demonstrated from additional perspectives—for example, through target-slice performance, robustness across different target attributes or failure slices, data efficiency at more constrained budgets, or a more detailed analysis of which samples are selected and why they lead to better repair. Such evidence could more clearly establish the benefit of the proposed selection mechanism beyond the small improvement in overall accuracy.

Thank you again for the careful and comprehensive response.

Replying to Official Comment by Reviewer hG6j

Response to Reviewer hG6j — Discussion Follow-Up (Part 1)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

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Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer hG6j

Revisions (/revisions?id=j229GzamV5)

Comment:

Dear Reviewer, thank you for the follow-up.

This response is organized into 5 parts.

Q1. Reusable Source-Attribute Metadata

Thank you for recognizing the distinction between the completeness of the cost evaluation and the practical value of the method. We would like to clarify that our previous end-to-end tables used a deliberately conservative first-run accounting: **every AttriSelect row included the full source-pool attribute-assignment cost**, even though this preprocessing does not need to be repeated at each source ratio or repair iteration.

For a fixed task-level attribute vocabulary, source-pool assignment is a one-time preprocessing step. The resulting metadata can be reused across source ratios, attribute-aware selectors, and later repair iterations on the same task. The full source pool therefore does not need to be annotated again for every repair. The corrected tables below remove this reusable assignment component from the AttriSelect totals; all baseline times are unchanged.

KITTI wall-clock time after reusing source-attribute metadata.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	AttriSelect
0.1	42.17 min	42.18 min	48.82 min	1.37 h	1.12 h	43.47 min	43.53 min	42.20 min
0.3	43.84 min	43.85 min	50.28 min	1.44 h	1.20 h	45.08 min	45.63 min	43.87 min
0.5	45.18 min	45.18 min	51.73 min	1.49 h	1.27 h	46.42 min	47.47 min	45.20 min
0.7	47.04 min	47.05 min	53.46 min	1.56 h	1.35 h	48.16 min	49.73 min	47.07 min
0.9	48.86 min	48.87 min	55.53 min	1.63 h	1.42 h	50.34 min	51.46 min	48.89 min

COCO wall-clock time after reusing source-attribute metadata.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	AttriSelect
0.1	4.41 min	4.42 min	10.98 min	16.18 min	19.63 min	5.08 min	6.97 min	4.44 min
0.3	8.18 min	8.18 min	14.74 min	39.86 min	49.76 min	8.93 min	17.86 min	8.20 min

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	AttriSelect
0.5	12.37 min	12.36 min	19.04 min	1.08 h	1.56 h	13.08 min	29.47 min	12.39 min
0.7	15.73 min	15.75 min	22.27 min	1.45 h	2.29 h	16.34 min	37.63 min	15.77 min
0.9	19.66 min	19.65 min	26.28 min	1.87 h	3.01 h	20.18 min	40.94 min	19.68 min

These adjusted tables represent later ratios or repair iterations after the task-level metadata has been created once. **`With metadata reuse, AttriSelect has nearly the same wall-clock cost as Random and Stratified at every ratio, while remaining substantially faster than CoreSet and BADGE.`** The earlier tables remain the conservative upper bound for a first repair on a task with no existing metadata.

Q2. Same-Information Repair Baselines Across Downstream Tasks

Response: We extended the same-information comparison across all three downstream tasks considered in this follow-up: ImageNet10 classification, COCO pose estimation, and KITTI object detection. For each task, all methods share the source pool, VLM attributes, diagnosed slice, fixed 1,000-image generated repair set, source ratios, diagnosis checkpoint, five seeds (0, 42, 1234, 3407, and 2024), and repair schedule. **Only the source-subset selection rule changes.**

In addition to the three direct attribute-aware controls—Target-Slice + Stratified, Attribute-Filtered Random, and Attribute-Filtered Entropy—we added a stronger **Generated-Complement** baseline. It uses the same generated-set attribute statistics and source metadata as AttriSelect, prioritizes source samples whose attributes are underrepresented in the generated set, and maintains task-specific class or proxy-label balance. The full score and deterministic balancing procedure are described in our response to Reviewer yyPZ under **Q2. Generated-Complement baseline**. We also include all seven generic baselines from the manuscript appendix: Random, Stratified, Entropy, CoreSet, BADGE, UHerding, and PPAL.

ImageNet10 same-information repair comparison. Metric: FR (%). Each entry is the five-seed mean ± standard deviation. Attribute-aware baselines are marked with inline backticks and rendered in red on OpenReview. Ratio Mean averages the five FR means at source ratios 0.1/0.3/0.5/0.7/0.9; standard deviations are not pooled.

Replying to Official Comment by Reviewer hG6j

Response to Reviewer hG6j — Discussion Follow-Up (Part 2)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

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Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer hG6j

Revisions (/revisions?id=yfZkt5wlR7)

Comment:

Q2. Same-Information Repair Baselines Across Downstream Tasks (continued)

ImageNet10.

Source Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	`Target-Slice + Stratified`	`Attribute-Filtered Random`	`Attribute-Filtered Entropy`	`Generated-Complement`	`AttriSelect`
0.1	3.40 ± 0.88	3.47 ± 1.31	2.87 ± 0.57	3.40 ± 1.13	5.31 ± 0.78	5.59 ± 0.81	3.96 ± 0.92	4.60 ± 1.80	5.91 ± 1.20	5.09 ± 0.88	1.50 ± 0.87	**6.93 ± 1.52**
0.3	4.78 ± 0.71	6.93 ± 1.27	4.21 ± 0.57	6.93 ± 0.42	8.84 ± 1.27	10.47 ± 1.52	7.75 ± 0.78	7.50 ± 0.85	7.39 ± 1.49	6.79 ± 0.81	5.31 ± 1.04	**11.28 ± 1.56**
0.5	7.32 ± 1.06	9.66 ± 0.60	5.59 ± 0.96	8.03 ± 0.60	10.47 ± 1.03	8.38 ± 0.99	8.84 ± 0.57	9.37 ± 1.31	9.52 ± 1.34	9.52 ± 1.10	11.29 ± 0.76	**11.85 ± 0.35**
0.7	8.84 ± 1.52	6.12 ± 1.52	6.40 ± 1.49	7.75 ± 1.52	10.75 ± 0.39	10.47 ± 1.20	8.84 ± 1.20	9.23 ± 2.30	10.65 ± 1.34	10.93 ± 0.71	8.57 ± 1.18	**12.10 ± 1.31**
0.9	10.22 ± 1.41	10.47 ± 0.39	10.22 ± 0.92	6.12 ± 1.24	9.37 ± 0.96	11.57 ± 1.17	11.85 ± 1.45	11.04 ± 0.92	10.47 ± 1.27	11.67 ± 1.59	10.48 ± 0.95	**13.76 ± 0.67**
Ratio Mean	6.91	7.33	5.86	6.44	8.95	9.30	8.25	8.35	8.79	8.80	7.43	**11.19**

COCO pose estimation. Metric: Fix Rate (FR, %).

Source Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Target-Slice + Stratified	Attribute-Filtered Random	Attribute-Filtered Entropy	Generated-Complement	AttriSelect
0.1	3.56 ± 0.68	5.08 ± 0.78	5.61 ± 0.46	-0.61 ± 0.98	2.03 ± 0.95	2.71 ± 1.07	1.17 ± 0.37	5.63 ± 0.66	-3.42 ± 0.87	4.47 ± 0.91	2.50 ± 0.83	**6.86 ± 1.07**
0.3	3.37 ± 0.73	8.32 ± 0.51	7.64 ± 0.71	8.64 ± 0.76	6.12 ± 1.07	4.32 ± 0.61	6.42 ± 0.49	9.38 ± 0.62	6.53 ± 0.84	8.37 ± 0.57	7.22 ± 0.68	**9.86 ± 0.54**
0.5	7.49 ± 0.49	8.00 ± 0.66	9.27 ± 0.63	6.47 ± 0.88	7.42 ± 0.44	6.12 ± 0.93	6.83 ± 0.54	8.83 ± 0.74	7.11 ± 0.76	8.16 ± 0.35	8.67 ± 0.79	**10.13 ± 0.29**
0.7	6.08 ± 1.00	9.69 ± 0.51	9.54 ± 0.54	9.10 ± 0.54	8.64 ± 0.39	6.17 ± 1.00	7.83 ± 1.07	9.24 ± 0.81	9.26 ± 0.68	9.68 ± 0.64	9.56 ± 0.72	**10.18 ± 0.59**
0.9	4.61 ± 0.98	9.49 ± 0.51	6.54 ± 0.44	7.32 ± 0.56	7.59 ± 0.68	7.05 ± 0.34	9.74 ± 0.61	8.68 ± 0.69	6.31 ± 0.83	9.78 ± 0.79	6.96 ± 0.91	**11.05 ± 0.76**
Ratio Mean	5.02	8.12	7.72	6.18	6.36	5.27	6.40	8.35	5.16	8.09	6.98	**9.62**

KITTI object detection. Metric: Fix Rate (FR, %).

Source Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Target-Slice + Stratified	Attribute-Filtered Random	Attribute-Filtered Entropy	Generated-Complement	AttriSelect
0.1	-9.65 ± 0.48	-7.27 ± 0.85	-7.52 ± 0.85	-8.89 ± 0.89	-6.95 ± 0.57	-9.24 ± 0.71	-8.32 ± 0.82	-8.24 ± 0.77	-6.14 ± 0.68	-6.81 ± 0.59	-6.24 ± 0.84	**-5.81 ± 0.53**
0.3	-0.48 ± 0.39	-1.23 ± 0.82	0.23 ± 0.75	-0.32 ± 0.30	-0.05 ± 0.69	-1.21 ± 0.48	-1.33 ± 0.89	-0.62 ± 0.56	0.04 ± 0.74	0.13 ± 0.46	0.17 ± 0.43	**0.46 ± 0.53**
0.5	2.13 ± 0.69	1.23 ± 0.89	3.20 ± 1.01	3.04 ± 0.98	2.29 ± 0.30	1.12 ± 0.62	2.17 ± 0.48	2.83 ± 0.36	2.58 ± 0.91	3.25 ± 0.33	1.72 ± 0.58	**3.36 ± 0.27**
0.7	3.20 ± 0.89	3.06 ± 0.30	3.34 ± 0.55	3.34 ± 0.30	2.08 ± 0.39	3.25 ± 0.55	3.57 ± 0.69	3.54 ± 0.48	2.93 ± 0.47	3.43 ± 0.61	**4.06 ± 0.71**	3.52 ± 0.89
0.9	4.37 ± 0.89	4.66 ± 0.85	4.66 ± 0.96	3.54 ± 0.25	4.21 ± 0.41	4.85 ± 1.01	3.89 ± 0.59	4.38 ± 0.82	4.46 ± 0.31	4.68 ± 0.88	3.74 ± 0.61	**5.12 ± 0.27**
Ratio Mean	-0.09	0.09	0.78	0.14	0.32	-0.25	0.00	0.38	0.77	0.94	0.69	**1.33**

Replying to Official Comment by Reviewer hG6j

Response to Reviewer hG6j — Discussion Follow-Up (Part 3)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

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Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer hG6j

Revisions (/revisions?id=v0PayIlsHV)

Comment:

Q2. Same-Information Repair Baselines Across Downstream Tasks (continued)

Takeaway. The ImageNet10 controls show that repair-relevant attributes help simple selectors. Attribute-Filtered Random and Attribute-Filtered Entropy beat their generic counterparts at all five ratios, and Target-Slice + Stratified beats Stratified at four of five. Attribute access by itself is not enough, however. **AttriSelect has the highest Ratio Mean on ImageNet10, COCO, and KITTI, and the highest FR at 14 of the 15 task--ratio points.** The exception is KITTI at ratio 0.7, where Generated-Complement is higher by 0.54 FR points. **`Under matched information and repair budgets, AttriSelect still gives the best aggregate performance across classification, pose estimation, and object detection.`**

Stability under extreme source budgets. Budget size affects both traditional Active Learning methods and the same-information baselines. At KITTI ratio 0.1, only 500 source images are selected. AttriSelect has the highest FR, while the other methods have larger drops. COCO ratio 0.9 selects 18,000 source images. Several methods are worse than at ratio 0.7, but AttriSelect rises from 10.18 to 11.05 FR. **`AttriSelect is the only evaluated method whose mean FR increases at every successive ratio on all three tasks.`**

Q3. More Constrained Source Budgets

We further tested more constrained source budgets on ImageNet10 and COCO. The selected source sets contain only 208/520 ImageNet10 images and 400/1,000 COCO images at source ratios 0.02/0.05. All methods use the same diagnosis and repair settings. The metric is Fix Rate (FR, %); entries are mean ± standard deviation, and Ratio Mean averages the two displayed FR means without pooling their standard deviations.

ImageNet10 classification under constrained source budgets. Metric: FR (%; mean ± standard deviation).

Source Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Generated–Complement	Target–Slice + Stratified	Attribute–Filtered Random	Attribute–Filtered Entropy	AttriSelect
0.02	-3.10 ± 1.22	-2.68 ± 1.08	-2.35 ± 1.34	-3.42 ± 1.16	-2.12 ± 1.27	-2.57 ± 1.05	-2.89 ± 1.18	-1.77 ± 1.21	-2.04 ± 1.38	0.68 ± 0.94	-0.95 ± 1.07	**1.23 ± 0.72**
0.05	1.84 ± 0.91	2.41 ± 0.84	2.76 ± 1.02	1.58 ± 0.88	2.91 ± 0.95	2.64 ± 0.90	2.20 ± 0.82	3.13 ± 0.86	3.95 ± 0.92	3.13 ± 0.78	2.32 ± 0.81	**5.58 ± 0.61**
Ratio Mean	-0.63	-0.14	0.21	-0.92	0.40	0.04	-0.35	0.68	0.96	1.91	0.69	**3.41**

COCO pose estimation under constrained source budgets. Metric: FR (%; mean ± standard deviation).

Source Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Generated–Complement	Target–Slice + Stratified	Attribute–Filtered Random	Attribute–Filtered Entropy	AttriSelect
0.02	-8.75 ± 1.42	-7.15 ± 1.31	-5.62 ± 1.18	-9.30 ± 1.57	-6.40 ± 1.26	-8.10 ± 1.49	-7.00 ± 1.35	-2.84 ± 1.16	-7.80 ± 1.43	-2.46 ± 1.02	-6.23 ± 1.28	**-2.02 ± 0.91**
0.05	-4.20 ± 1.12	-2.96 ± 0.98	-1.84 ± 0.87	-5.10 ± 1.24	-2.55 ± 0.93	-4.00 ± 1.15	-3.20 ± 1.01	-31.00 ± 2.34	-11.64 ± 1.76	-0.61 ± 0.74	-1.10 ± 0.82	**1.63 ± 0.68**
Ratio Mean	-6.48	-5.06	-3.73	-7.20	-4.48	-6.05	-5.10	-16.92	-9.72	-1.54	-3.67	**-0.20**

Takeaway. At these budgets, many baselines have negative FR: with only 208--520 ImageNet10 images or 400--1,000 COCO images, the fixed repair schedule can make the model worse rather than better. This suggests that extremely small source budgets may require a different source/generated mixture or a longer repair schedule. We intentionally keep the current settings unchanged so that the selector remains the only comparison variable. **`Even in this constrained regime, AttriSelect is best at all four dataset--ratio points and has the highest Ratio Mean on both tasks.`** Its Ratio Mean exceeds the strongest alternative by **1.50 FR points on ImageNet10** and **1.34 FR points on COCO**. The absolute gains are modest because the repair data and training budget are small, but the ordering is consistent under the same controlled protocol.

Replying to Official Comment by Reviewer hG6j

Response to Reviewer hG6j — Discussion Follow-Up (Part 4)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

03 Aug 2026, 02:54 (modified: 03 Aug 2026, 14:22)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer hG6j

Revisions (/revisions?id=xgl4ec9zXO)

Comment:

Q4. Exact-Slice and Selected-Sample Evidence

The cross-task results in Q2 cover classification, pose estimation, and object detection under task-specific diagnosed failures. We also evaluated the exact diagnosed conjunctions at ratio 0.5. On ImageNet10, AttriSelect improves the exact "pink AND fabric" slice by **13.07 percentage points** and its non-target complement by **3.23 points**. On COCO, it improves the exact "brown AND rainy" slice by **10.13 points** and the non-target complement by **4.04 points**. The full five-seed tables, slice sizes, and variability are reported in our response to Reviewer yyPZ under **Q3. Exact diagnosed-slice and non-target performance**. The positive target and non-target changes show that the overall gains include the diagnosed failures without reducing performance on the remaining test data.

To examine which source samples are selected, we compare AttriSelect with Generated-Complement and the three direct same-information controls on ImageNet10. All methods use the same source pool, fixed 1,000-image generated set, diagnosis checkpoint, source ratios, and 15-epoch repair schedule. Downstream FR values are five-seed means.

Each `# (%)` entry gives the sample count and its percentage within the selected source subset. Exact Target matches "pink AND fabric"; Single-Factor matches exactly one of pink or fabric; Background matches neither. Unique Class--Attribute Tuples counts distinct (class, color, background, texture) combinations. Classes Covered is the number of ImageNet10 classes represented in the selected subset. FR is Relative Error Reduction, $(M_{repair} - M_{diagnosis})/(1 - M_{diagnosis}) \times 100$; higher is better.

Ratio	Method	Exact Target # (%)	Single-Factor # (%)	Background # (%)	Unique Class--Attribute Tuples	Classes Covered	FR
0.1	Target-Slice + Stratified	130 (12.50%)	141 (13.56%)	769 (73.94%)	382	10	4.60
0.1	Attribute-Filtered Random	30 (2.88%)	686 (65.96%)	324 (31.15%)	401	10	5.91
0.1	Attribute-Filtered Entropy	34 (3.27%)	682 (65.58%)	324 (31.15%)	410	10	5.09
0.1	Generated-Complement	0 (0.00%)	58 (5.58%)	982 (94.42%)	396	10	1.50
0.1	AttriSelect	24 (2.31%)	269 (25.87%)	747 (71.83%)	680	10	6.93
0.3	Target-Slice + Stratified	130 (4.17%)	581 (18.62%)	2409 (77.21%)	762	10	7.50
0.3	Attribute-Filtered Random	60 (1.92%)	1130 (36.22%)	1930 (61.86%)	780	10	7.39
0.3	Attribute-Filtered Entropy	70 (2.24%)	1120 (35.90%)	1930 (61.86%)	801	10	6.79
0.3	Generated-Complement	0 (0.00%)	266 (8.53%)	2854 (91.47%)	692	10	5.31
0.3	AttriSelect	40 (1.28%)	823 (26.38%)	2257 (72.34%)	1165	10	11.28
0.5	Target-Slice + Stratified	130 (2.50%)	1005 (19.33%)	4065 (78.17%)	1007	10	9.37
0.5	Attribute-Filtered Random	101 (1.94%)	1505 (28.94%)	3594 (69.12%)	989	10	9.52
0.5	Attribute-Filtered Entropy	92 (1.77%)	1514 (29.12%)	3594 (69.12%)	995	10	9.52
0.5	Generated-Complement	0 (0.00%)	618 (11.88%)	4582 (88.12%)	902	10	11.29
0.5	AttriSelect	95 (1.83%)	1183 (22.75%)	3922 (75.42%)	1318	10	11.85
0.7	Target-Slice + Stratified	130 (1.79%)	1449 (19.91%)	5700 (78.31%)	1194	10	9.23
0.7	Attribute-Filtered Random	125 (1.72%)	1834 (25.20%)	5320 (73.09%)	1186	10	10.65
0.7	Attribute-Filtered Entropy	118 (1.62%)	1841 (25.29%)	5320 (73.09%)	1168	10	10.93
0.7	Generated-Complement	0 (0.00%)	1034 (14.21%)	6245 (85.79%)	1044	10	8.57
0.7	AttriSelect	95 (1.31%)	1519 (20.87%)	5665 (77.83%)	1379	10	12.10
0.9	Target-Slice + Stratified	130 (1.39%)	1884 (20.13%)	7346 (78.48%)	1345	10	11.04

Ratio	Method	Exact Target # (%)	Single-Factor # (%)	Background # (%)	Unique Class--Attribute Tuples	Classes Covered	FR
0.9	Attribute-Filtered Random	129 (1.38%)	2038 (21.77%)	7193 (76.85%)	1325	10	10.47
0.9	Attribute-Filtered Entropy	129 (1.38%)	2038 (21.77%)	7193 (76.85%)	1315	10	11.67
0.9	Generated-Complement	0 (0.00%)	1493 (15.95%)	7867 (84.05%)	1187	10	10.48
0.9	AttriSelect	130 (1.39%)	1888 (20.17%)	7342 (78.44%)	1403	10	13.76

Replying to Response to Reviewer hG6j — Discussion Follow-Up (Part 4)

Response to Reviewer hG6j — Discussion Follow-Up (Part 5)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

03 Aug 2026, 02:55 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer hG6j

Comment:

Q4. Exact-Slice and Selected-Sample Evidence (continued)

Class Imbalance is the difference between the largest and smallest selected counts across the ten ImageNet10 classes; lower is more balanced. Generated Redundancy is the selected-set mean fraction of the two generated-supported target fields matched by each source sample, namely "color=pink" and "texture=fabric"; lower means less duplication of the generated target attributes. Jaccard with AttriSelect is $|S_m \cap S_A|/|S_m \cup S_A|$, where S_m and S_A are the source subsets selected by method m and AttriSelect at the same ratio. Mean Pairwise Attribute Distance is the mean categorical Hamming distance over color, background, and texture across all unordered selected-source pairs; higher indicates broader attribute variation.

Ratio	Method	Class Imbalance ↓	Generated Redundancy ↓	Jaccard with AttriSelect	Mean Pairwise Attribute Distance ↑	FR
0.1	Target-Slice + Stratified	0	0.1928	0.0500	0.7870	4.60
0.1	Attribute-Filtered Random	0	0.3587	0.1129	0.7188	5.91
0.1	Attribute-Filtered Entropy	0	0.3606	0.1141	0.7265	5.09
0.1	Generated-Complement	0	0.0279	0.0711	0.7853	1.50
0.1	AttriSelect	0	0.1524	1.0000	0.8597	6.93
0.3	Target-Slice + Stratified	0	0.1348	0.1782	0.7902	7.50
0.3	Attribute-Filtered Random	0	0.2003	0.2086	0.7772	7.39
0.3	Attribute-Filtered Entropy	0	0.2019	0.2197	0.7805	6.79
0.3	Generated-Complement	0	0.0426	0.1710	0.7861	5.31
0.3	AttriSelect	2	0.1447	1.0000	0.8414	11.28
0.5	Target-Slice + Stratified	0	0.1216	0.3281	0.7912	9.37
0.5	Attribute-Filtered Random	0	0.1641	0.3437	0.7847	9.52
0.5	Attribute-Filtered Entropy	0	0.1633	0.3437	0.7861	9.52
0.5	Generated-Complement	0	0.0594	0.3161	0.7881	11.29
0.5	AttriSelect	4	0.1320	1.0000	0.8272	11.85
0.7	Target-Slice + Stratified	1	0.1174	0.5391	0.7919	9.23

Ratio	Method	Class Imbalance ↓	Generated Redundancy ↓	Jaccard with AttriSelect	Mean Pairwise Attribute Distance ↑	FR
0.7	Attribute-Filtered Random	1	0.1432	0.5323	0.7883	10.65
0.7	Attribute-Filtered Entropy	1	0.1427	0.5300	0.7879	10.93
0.7	Generated-Complement	1	0.0710	0.5165	0.7896	8.57
0.7	AttriSelect	5	0.1174	1.0000	0.8124	12.10
0.9	Target-Slice + Stratified	0	0.1145	0.8169	0.7915	11.04
0.9	Attribute-Filtered Random	0	0.1226	0.8154	0.7904	10.47
0.9	Attribute-Filtered Entropy	0	0.1226	0.8162	0.7909	11.67
0.9	Generated-Complement	0	0.0798	0.8080	0.7899	10.48
0.9	AttriSelect	6	0.1147	1.0000	0.7977	13.76

At the more constrained ratios, each Best-Control value is selected independently from Target-Slice + Stratified, Attribute-Filtered Random, Attribute-Filtered Entropy, and Generated-Complement. Best-Control is the maximum for FR, Unique Tuples, and Attribute Distance and the minimum for Class Imbalance. Each margin or gain is the AttriSelect value minus the corresponding Best-Control value.

Ratio	AttriSelect FR	Best-Control FR	FR Margin	AttriSelect Class Imbalance	Best-Control Class Imbalance	AttriSelect Unique Tuples	Best-Control Unique Tuples	Tuple Gain	AttriSelect Attribute Distance	Best-Control Attribute Distance	Distance Gain
0.02	1.23	0.68	+0.55	1	1	184	142	+42	0.8823	0.7695	+0.1128
0.05	5.58	3.95	+1.63	0	0	397	267	+130	0.8680	0.7828	+0.0853

Takeaway. At ratios 0.02 and 0.05, AttriSelect improves five-seed FR over the strongest control by 0.55 and 1.63 points. It also covers 42 and 130 more class--attribute tuples and increases Mean Pairwise Attribute Distance by 0.1128 and 0.0853, while matching the best Class Imbalance. **`With only 208 or 520 selected source samples, AttriSelect has the highest FR and covers a broader range of classes and attributes.`**
Across ratios 0.1--0.9, AttriSelect also has the highest FR, Unique Class--Attribute Tuples, and Mean Pairwise Attribute Distance among the matched controls. Generated-Complement has lower Generated Redundancy but lower FR, so avoiding generated-target duplication alone is insufficient. This pattern is consistent with AttriSelect retaining repair-relevant samples without narrowing class and attribute coverage.

Official Review of Submission12417 by Reviewer KaLa

Official Review by Reviewer KaLa 15 Jun 2026, 03:38 (modified: 04 Aug 2026, 08:23)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer KaLa Revisions (/revisions?id=12XbHqOFYG)

Summary:
This paper studies targeted vision model repair from the perspective of repair-time source-data selection. Given a diagnosed failure slice and generated repair images, the proposed AttriSelect method selects a compact subset of original training examples to accompany the generated data during retraining, aiming to reduce computational cost while preserving repair effectiveness. The method uses attribute-driven criteria, including class-conditional rarity, generated-data coverage control, and soft class balance, to better capture repair-relevant semantic slices than conventional uncertainty-, diversity-, or gradient-based active learning methods. Experiments across image classification, pose estimation, and object detection suggest the effectiveness of AttriSelect.

Contribution Type: General: Most submissions will fall into this type.
Strengths And Weaknesses:
Strengths.

1. The paper presents an interesting and inspiring perspective by formulating data selection as an important component of the targeted model repair process.
2. The experimental results are promising and suggest that the proposed attribute-driven selection strategy can improve repair performance across different vision tasks.

Weaknesses.

1. The problem formulation needs to be clarified. It is not always clear whether the proposed method is selecting generated repair data, original source data, or a combination of both. The descriptions in the abstract, introduction, and conclusion appear somewhat vague or potentially inconsistent on this point. The authors are encouraged to state explicitly what data pool AttriSelect operates on and how the selected data are used during retraining.
2. The categorization of prior model repair methods requires further clarification. In particular, reference [7] seems to be associated with both the first and second lines of work, which makes the taxonomy confusing. The authors should revise this discussion to make the distinctions among different categories more precise.
3. It is unclear what repair method or methods are used together with AttriSelect in the experiments. Since AttriSelect is a data selection strategy rather than a complete repair algorithm, the experimental section should clearly specify the underlying repair pipeline, retraining/fine-tuning procedure, and how the selected source samples are combined with the generated repair images.
4. The paper states that AttriSelect is best viewed as one component in a localized repair pipeline. However, the experiments do not appear to evaluate its compatibility with multiple repair pipelines. It would strengthen the paper if the authors could test AttriSelect with different repair methods or, at least, discuss the extent to which the conclusions depend on the specific repair pipeline used in the experiments.
5. The formatting of the references should be improved. For example, the names of journals and conferences should be made consistent, either using full names or standard abbreviations throughout the reference list.

Quality: 3: good
Clarity: 3: good
Significance: 3: good
Originality: 3: good
Questions:

1. Could the authors clearly specify whether AttriSelect selects from the original source/training data, the generated repair data, or both at related parts?
2. Since AttriSelect is presented as a data selection component rather than a complete repair method, could the authors specify the exact repair pipeline used in each experiment?
3. The paper states that AttriSelect is best viewed as one component in a localized repair pipeline. Could the authors provide evidence or discussion about whether AttriSelect remains effective when paired with different repair pipelines or retraining strategies?
4. Could the authors improve the categorization of prior repair methods?

Limitations:
yes

Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
NA

Code Of Conduct Acknowledgement: Yes
Responsible Reviewing Acknowledgement: Yes

Response to Reviewer KaLa (Part 1)

Official Comment
by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))
30 Jul 2026, 01:56 (modified: 30 Jul 2026, 03:05)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer KaLa
Revisions (/revisions?id=nlvojL88SP)

Comment:
Dear Reviewer KaLa, thank you for asking us to clarify the selected data pool, repair protocol, and compatibility with other pipelines.

This response is organized into four parts.

W1/Q1 — Selected Data Pool

> "The problem formulation needs to be clarified. It is not always clear whether the proposed method is selecting generated repair data, original source data, or a combination of both. The descriptions in the abstract, introduction, and conclusion appear somewhat vague or potentially inconsistent on this point. The authors are encouraged to state explicitly what data pool AttriSelect operates on and how the selected data are used during retraining."

Response. `AttriSelect selects only from the original source or training pool. It does not select generated repair images or rank a joint source--generated pool.` Let r denote the source-selection ratio in our tables. AttriSelect returns $\mathcal{D}^{sel}$, a subset of $\mathcal{D}^{src}$. If the source pool contains N images, the selected subset contains $\lfloor rN \rfloor$ images. In every main experiment, the upstream repair pipeline diagnoses a failure slice and constructs a generated repair set. That set contains the same 1,000 images for every selector and source ratio. The retraining set $\mathcal{D}^{repair}$ is the union of the selected source subset $\mathcal{D}^{sel}$ and the fixed generated set $\mathcal{D}^{gen}$.

AttriSelect uses attribute counts from the fixed generated set to identify values that are already well represented and avoid redundant source examples. Every ranked candidate and every selected image still comes from the source pool. We then continue training or fine-tune the diagnosis checkpoint on $\mathcal{D}^{repair}$. Therefore, changing r changes only the number and composition of real source images; the generated-image budget and membership do not change.

We will use this terminology consistently in the abstract, introduction, problem formulation, framework figure, experimental protocol, and conclusion: the upstream pipeline supplies the diagnosed slice and fixed generated set; AttriSelect selects a compact source subset; retraining uses their union. We will remove any notation that suggests a selected generated subset.

W2/Q4 — Prior-Work Taxonomy

> "The categorization of prior model repair methods requires further clarification. In particular, reference [7] seems to be associated with both the first and second lines of work, which makes the taxonomy confusing. The authors should revise this discussion to make the distinctions among different categories more precise."

Response. `The repeated citation is intentional because HiBug participates in two stages: it discovers attribute–defined failure slices and generates repair examples for them.`

We will organize the revision by pipeline stage rather than present the categories as mutually exclusive. The stages are **failure-slice discovery, repair-data construction, source-data selection, and model retraining**. HiBug can then appear in more than one stage without making the taxonomy inconsistent.

Reference [7] is HiBug, an early method that discovers human-interpretable model bugs and supports targeted repair with generated examples in computer-vision tasks. All submitted experiments place our selector inside the SafeFix repair pipeline. SafeFix addresses two limitations of direct generative augmentation: **Semantic Drift**, in which attributes unrelated to the target edit change, and **Distribution Mismatch**, in which synthetic images depart from the source-data distribution. It reduces these problems through controlled editing of real training images and VLM-based filtering.

`W2/Q4 addresses the prior–work taxonomy; W4/Q3 reports the new five–seed HiBug experiment on compatibility with another repair pipeline.`

Replying to Response to Reviewer KaLa (Part 1)

Response to Reviewer KaLa (Part 2)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

30 Jul 2026, 01:56 (modified: 30 Jul 2026, 03:06)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer KaLa

Revisions (/revisions?id=oTIizjQWoJ)

Comment:

W3/Q2 — Repair-Pipeline Implementation

> "It is unclear what repair method or methods are used together with AttriSelect in the experiments. Since AttriSelect is a data selection strategy rather than a complete repair algorithm, the experimental section should clearly specify the underlying repair pipeline, retraining/fine-tuning procedure, and how the selected source samples are combined with the generated repair images."

Response. `All submitted experiments place AttriSelect inside the SafeFix repair pipeline.` SafeFix follows the targeted-generation direction introduced by HiBug, using controlled edits of real source images and VLM-based filtering to construct repair images. AttriSelect leaves diagnosis and generation unchanged. It replaces only the rule for choosing the real source examples that accompany the generated repair images.

The implemented workflow is:

Repair-pipeline summary.
Upstream SafeFix pipeline
Diagnosis checkpoint θ_t
↓
Diagnose an attribute-defined failure slice T_{bug}
↓
SafeFix controlled generation using ControlNet/Stable Diffusion 1.5, soft HED conditioning, and 30 DDIM steps
↓
Fixed generated repair set $\mathcal{D}^{gen}$ (1,000 images)
AttriSelect source selection
Original source pool $\mathcal{D}^{src}$ with attribute metadata, together with attribute counts from $\mathcal{D}^{gen}$
↓
AttriSelect ranks source images only
↓
Selected source subset $\mathcal{D}^{sel}$
Repair training
Union of $\mathcal{D}^{sel}$ and $\mathcal{D}^{gen}$
↓
Matched repair training or fine-tuning of θ_t
↓
Repaired checkpoint θ_{t+1}

The generated images affect the AttriSelect score through their attribute counts, but they are never selection candidates. At source ratio r, if the source pool contains N images, the selector returns $\lfloor rN \rfloor$ images. The actual repair dataset $\mathcal{D}^{repair}$ is the union of the selected source subset $\mathcal{D}^{sel}$ and the fixed generated set $\mathcal{D}^{gen}$.

The appendix also gives the dataset-specific settings. For ImageNet10, we train the diagnosis-stage ResNet-18 for 30 epochs using cross-entropy loss, Adam, learning rate 1×10^{-4}, weight decay 1×10^{-3}, and batch size 64. Qwen3-VL supplies closed-set attribute assignments for the source images. Repair then runs for 15 epochs at learning rate 1×10^{-5}. For CelebA, COCO, and KITTI, repair uses half as many epochs as diagnosis and the same repair learning rate, 1×10^{-5}. Within each dataset-level comparison, all selectors use the same diagnosis checkpoint, failure slice, fixed 1,000-image generated set, source ratio, task model, and repair schedule. **`Only the source-subset selection rule changes.`**

We will move this workflow and these settings into the experimental section so that readers do not have to assemble the repair method, data composition, and training procedure from several appendix sections.

Replying to Response to Reviewer KaLa (Part 1)

Response to Reviewer KaLa (Part 3)

Official Comment
by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))
30 Jul 2026, 01:56 (modified: 30 Jul 2026, 03:06)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer KaLa
Revisions (/revisions?id=RfCHy0kEsQ)

Comment:

W4/Q3 — Compatibility with Alternative Pipelines

> "The paper states that AttriSelect is best viewed as one component in a localized repair pipeline. However, the experiments do not appear to evaluate its compatibility with multiple repair pipelines. It would strengthen the paper if the authors could test AttriSelect with different repair methods or, at least, discuss the extent to which the conclusions depend on the specific repair pipeline used in the experiments."

> Additional point in Q3: "Could the authors provide evidence or discussion about whether AttriSelect remains effective when paired with different [...] retraining strategies?"

Alternative Repair Pipeline: HiBug

Response. HiBug [3] is an attribute-based debugging and repair framework. It uses vision--language models to find interpretable failure slices and generate targeted repair examples. We test AttriSelect inside HiBug on CelebA, replacing the SafeFix-generated repair set with a HiBug-generated set. The diagnosis checkpoint, 80,000-image source pool, source ratios, generated budget, retraining configuration, and five seeds are matched across selectors. Every row uses the same fixed HiBug-generated set; only the source-selection rule changes.

CelebA with the HiBug repair pipeline and 80,000 training images. Metric: five-seed mean FR (%).

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours (HiBug)
0.1	4.59	2.33	6.92	-6.20	2.87	-1.36	5.58	**7.36**

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours (HiBug)
0.3	5.91	5.08	8.02	7.79	1.66	3.39	3.87	**8.07**
0.5	5.76	7.70	7.48	7.90	2.53	2.13	9.64	**12.54**
0.7	2.57	8.06	10.03	6.19	7.86	4.96	5.90	**11.16**
0.9	2.84	4.28	7.95	5.71	5.80	4.19	6.37	**13.42**

Key result. **`Under HiBug, AttriSelect is strongest at all five ratios, exceeding the best baseline by 0.44/0.05/2.90/1.13/5.47 FR points and by 2.00 points on average.`** These comparisons should be read within the HiBug table. Changing the upstream pipeline changes the generated repair set and can shift the absolute FR of every selector.

Alternative-Generator Evidence

Response. AttriSelect requires a diagnosed semantic slice, an annotated source pool, and a fixed generated repair set. To test dependence on the main ControlNet generator, the appendix substitutes InstructPix2Pix on CelebA and ICEdit on COCO. The selectors, generated-image budget, source ratios, and repair protocol remain fixed.

CelebA with InstructPix2Pix generation. Metric: FR (%).

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours (InstructPix2Pix)	Ours (ControlNet)
0.1	4.23	1.95	6.41	-6.65	2.34	-1.78	5.12	**7.65**	8.48
0.3	5.34	4.72	7.98	7.21	1.02	2.81	3.45	**8.12**	8.91
0.5	5.56	7.23	6.78	7.54	2.12	1.56	9.12	**11.17**	13.79
0.7	2.12	7.67	9.34	5.78	7.34	4.23	5.34	**12.13**	12.09
0.9	2.45	3.92	7.56	5.34	5.23	3.87	5.91	**13.87**	14.74

Key result. **`AttriSelect is the strongest source selector at every evaluated CelebA ratio under InstructPix2Pix, exceeding the strongest baseline by 2.51 FR points on average.`**

COCO pose estimation with ICEdit generation. Metric: FR (%).

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	Ours (ICEdit)	Ours (ControlNet)
0.1	3.78	5.23	5.87	-0.34	2.34	3.12	1.56	**7.12**	6.86
0.3	3.92	8.67	8.12	9.12	6.56	4.67	6.89	**10.34**	9.86
0.5	7.91	8.45	9.67	6.91	7.82	6.54	7.23	**10.67**	10.13
0.7	6.54	10.12	9.98	9.56	9.12	6.71	8.23	**10.87**	10.18
0.9	5.12	9.87	6.98	7.82	8.12	7.54	10.12	**11.56**	11.05

Key result. Under ICEdit, AttriSelect is strongest at every evaluated COCO ratio and exceeds the best baseline by **1.13 FR points on average**. **`Across the two generator substitutions, AttriSelect ranks first in all 10 evaluated dataset–ratio settings.`**

Replying to Response to Reviewer KaLa (Part 1)

Response to Reviewer KaLa (Part 4)

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

30 Jul 2026, 01:56 (modified: 30 Jul 2026, 03:07)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer KaLa

Revisions (/revisions?id=y5nbXoXLud)

Comment:

Scope of the Evidence

Scope. The empirical evidence covers one matched alternative repair pipeline (HiBug on CelebA) and two generator substitutions within the SafeFix-style workflow (InstructPix2Pix on CelebA and ICEdit on COCO).

The appendix theory gives conditions sufficient for transfer across repair pipelines and generators. It concerns source selection and does not depend on SafeFix, HiBug, a particular image generator, or a specific optimizer. Given a diagnosed target slice, an attribute-annotated source pool, and a generated repair set, the rarity-dominates-penalty result gives conditions under which target-slice source examples remain ranked above non-target examples after accounting for the generated-data distribution. Let B_c denote the selection

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budget allocated to class c, and let T_c denote the set of target-slice source examples in that class. The deterministic enrichment result then guarantees at least $\min(B_c, |T_c|)$ selected target-slice examples within class c, whereas an attribute-agnostic selector requires its uncertainty or coverage signal to be separately aligned with that slice.

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These guarantees apply before model updating, so the selected subset can feed checkpoint fine-tuning, continued training, or retraining from scratch. Under the appendix's repair-utility condition, the enrichment result gives expected repair utility at least as high as random or stratified selection under the same class allocation, with a strict advantage when the enrichment gap is strict. **Changing the repair pipeline or update rule affects the data counts and how the additional target support translates into the final metric; it does not remove the selection mechanism.** This is a conditional selection-level argument, not a guarantee for every optimizer, checkpoint, or data-weighting scheme.

W5 — Reference Formatting

> "The formatting of the references should be improved. For example, the names of journals and conferences should be made consistent, either using full names or standard abbreviations throughout the reference list."

Response. We standardized conference and journal names throughout the reference list using NeurIPS, ICLR, ICML, CVPR, ICCV, ECCV, WACV, AAAI, FAccT, and JMLR. We also corrected title and venue capitalization, protected proper names such as ImageNet, COCO, KITTI, and SafeFix, and fixed the malformed author field in the PPAL reference. The bibliography now uses one format.

Additional Evidence from Other Reviewer Responses

Several experiments in the other responses bear directly on KaLa's questions. The hG6j response gives every method the same ImageNet10 attributes and target-slice definition. AttriSelect leads at all five source ratios by 0.59 accuracy points on average over the best control.

The yyPZ response varies repair training from 15 to 60 epochs. AttriSelect remains strongest in all eight configurations. The Area Chair response reports source-only ImageNet10 experiments that train a new ResNet-18 rather than fine-tune a repair checkpoint. AttriSelect leads at all five one-shot budgets, and AttriSelect-AL improves the six-round Curve Mean by 1.40 points over BADGE.

The other responses report five-seed studies of slice granularity, alternative attribute sources, and incomplete vocabularies. **The observed advantage is therefore not confined to one information budget, repair schedule, checkpoint–update mode, slice definition, or attribute source.**

References

1. SafeFix: Targeted Model Repair via Controlled Image Generation. TMLR, 2026.
2. Actively Selecting Annotations Among Objects and Attributes. ICCV, 2011.
3. HiBug: On Human-Interpretable Model Debug. NeurIPS, 2023.

Official Comment by Reviewer KaLa

Official Comment by Reviewer KaLa 01 Aug 2026, 04:40

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer KaLa

Comment:

Thanks for the detailed rebuttal, which has addressed my major concerns regarding the formulation of the selection problem, the experimental setup, and the role of AttriSelect in the repair pipeline. I will raise my score.

Replying to Official Comment by Reviewer KaLa

Thanks to Reviewer KaLa

Official Comment

by Authors (RUIYU MAO (/profile?id=~RUIYU_MAO1), Yunhui Guo (/profile?id=~Yunhui_Guo2), Ouyang Xu (/profile?id=~Ouyang_Xu1), Baoming Zhang (/profile?id=~Baoming_Zhang2))

03 Aug 2026, 16:08 (modified: 03 Aug 2026, 17:29)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer KaLa

Revisions (/revisions?id=UuG0rWPDbi)

Comment:

Hi, thank you very much for your feedback! We are so glad that our previous responses clarified potential misunderstandings and addressed your concerns. We also sincerely appreciate your positive comments about the interesting topic and the comprehensive evaluation.

We have recently added several Discussion Follow-Up responses to the other reviewers. If you have a chance, you may find them by looking for our responses titled **"Discussion Follow-Up"** from the past day. We hope those additional discussions help further clarify our paper or answer any remaining questions you may have.

Thank you again for your time, your thoughtful feedback, and for considering our work.